南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

生物医学工程系
Department of Biomedical Engineering

Image Enhancement

Yiming Li
liym2019@sustech.edu.cn

2026/10/16



Lecture Roadmap

- 1. Choosing an intensity transformation
  Which intensity range needs more separation?

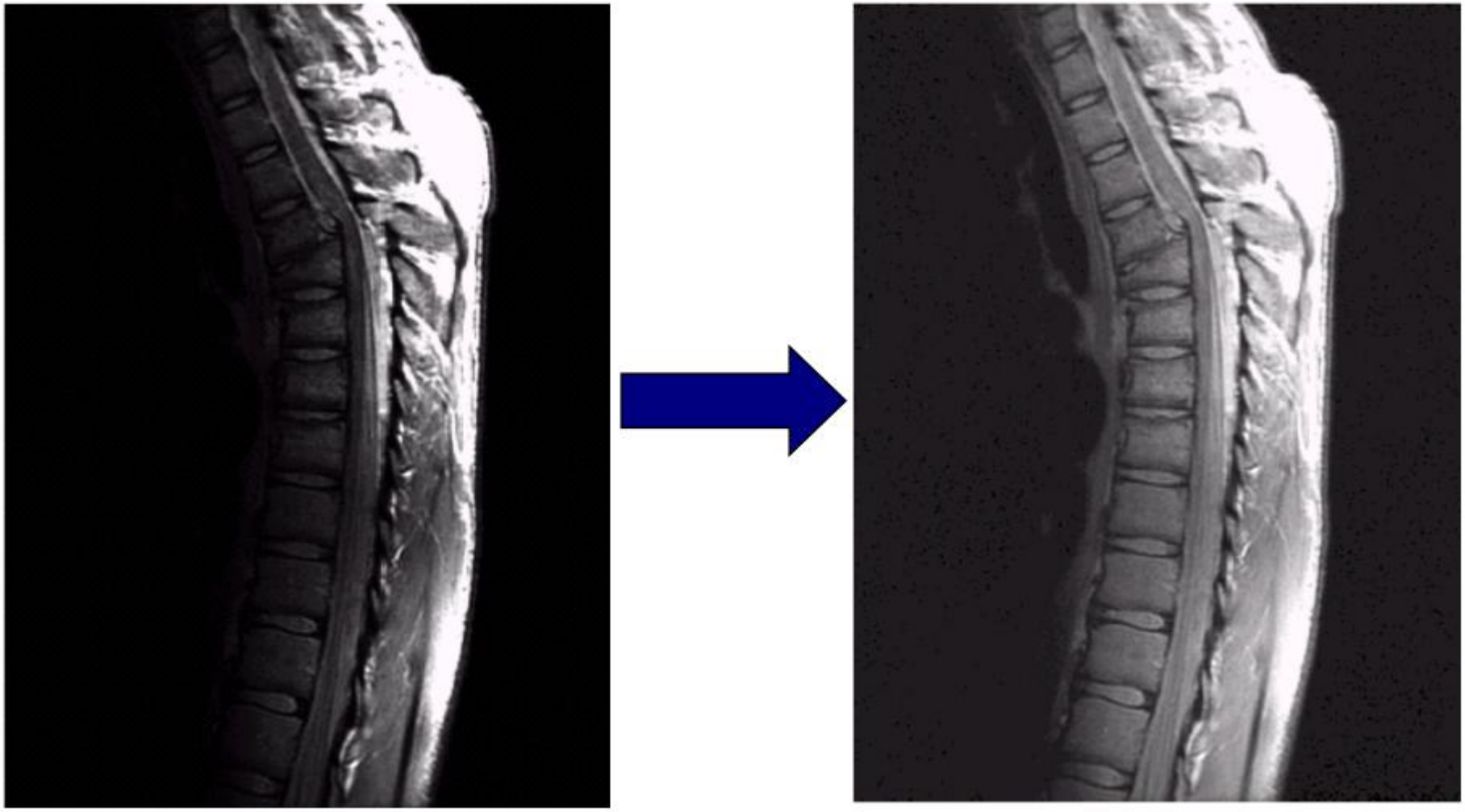
- 2. Designing a transformation from the histogram
  How can the image determine the mapping?

- 3. Beyond global enhancement
  When do we need a target distribution or local context?

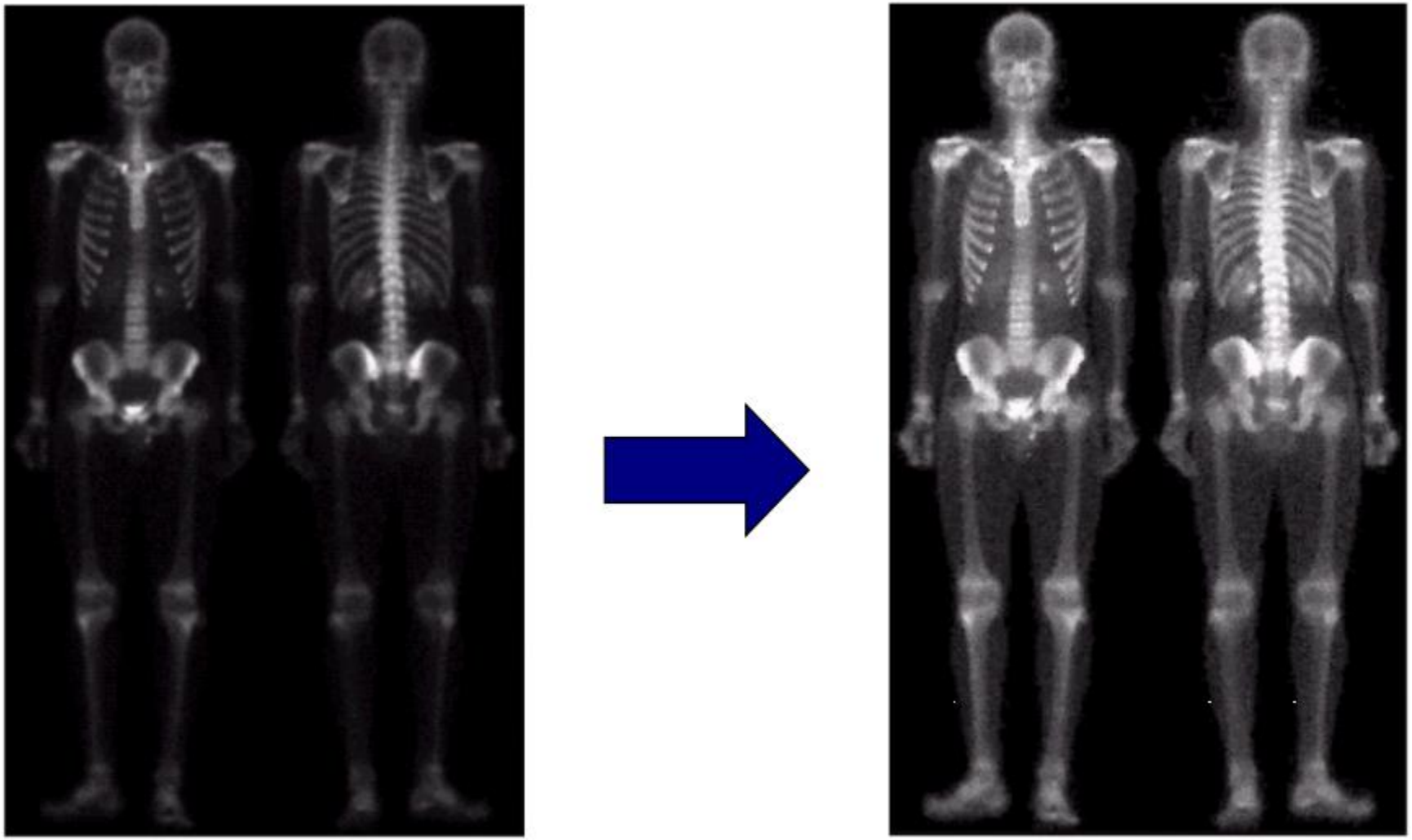
Purpose of Enhancement

- Enhancement makes task-relevant features easier to inspect.
- Possible goals: reveal detail, reduce noise, or improve appearance.
- Evaluation depends on the target region and the intended task.
- A brighter image does not necessarily reveal more useful detail.

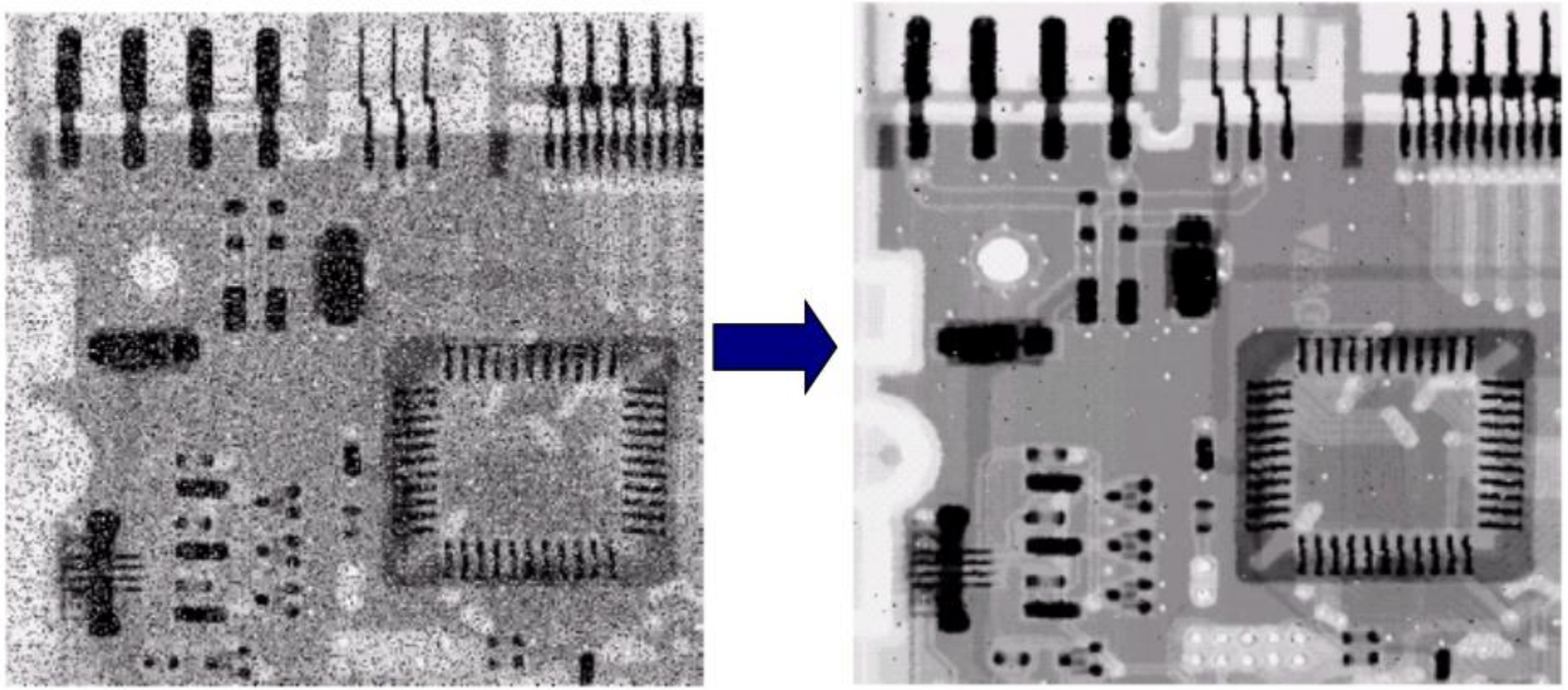
Example: Dark-Region Detail



Example: Improve Contrast



Example: Noise and Detail



Example: Skin Smoothing



Spatial and Frequency Domains

- There are two broad categories of image enhancement techniques
  - Spatial domain techniques
    - Direct manipulation of image pixels
  - Frequency domain techniques
    - Manipulation of Fourier transform or wavelet transform of an image

This part focuses on intensity transformations and histogram processing.



Intensity Ranges

- 8-bit unsigned images: $r \in [0, 255]$.
- General discrete images: $r \in \{0, \ldots, L - 1\}$.
- Normalized intensity: $u = r / (L - 1)$, $u \in [0, 1]$.
- State the range before applying a nonlinear transformation.

Spatial-Domain Operations

The general form of spatial domain processes

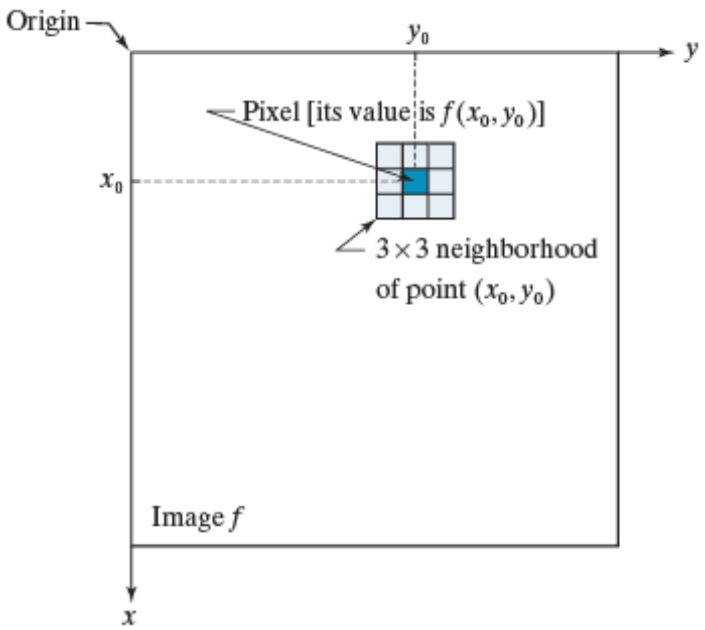
- Given the gray level of a pixel, use a function T to output a modified gray level

$$g(x,y) = T[f(x,y)]$$

$f(x,y)$: original _ image

$g(x,y)$: output _ image

$T[\bullet]$: trans _ function



The neighborhood size

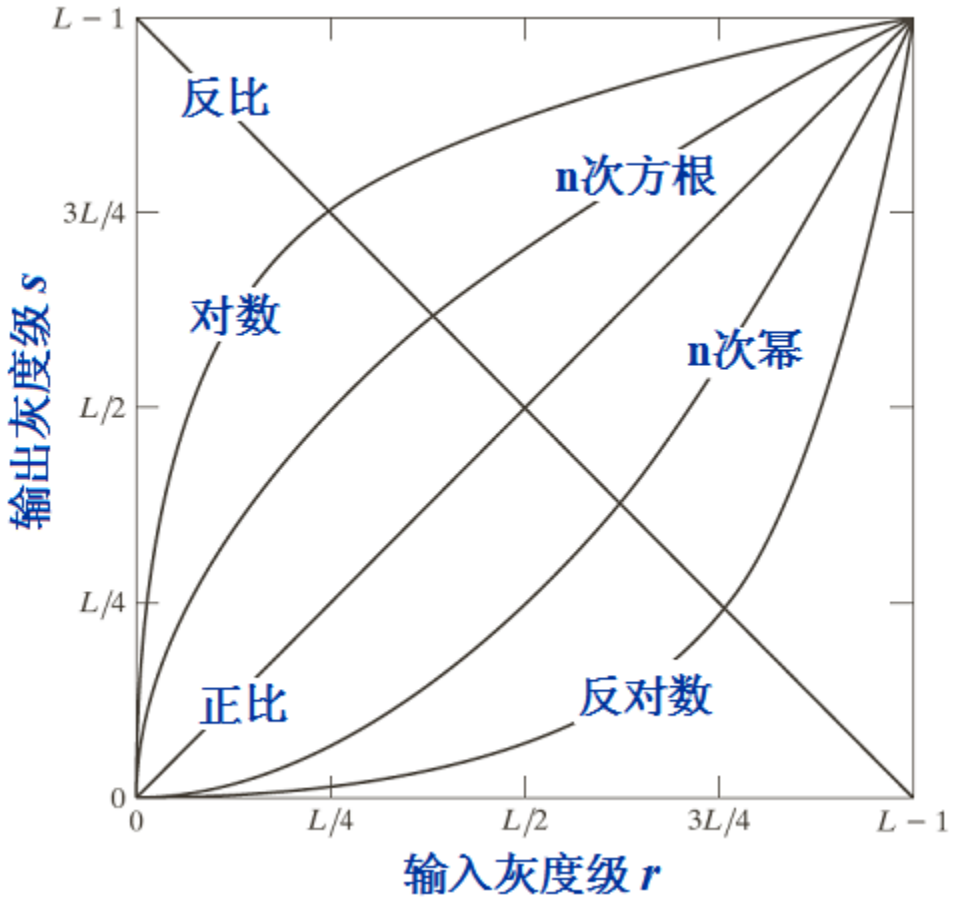
1. Smallest size 1×1: basic intensity transformation s = T(r)
2. n ×n neighborhood: spatial filtering
3. Statics of whole image: Enhancement based on histogram

Intensity Transformations

$$g(x,y) = T\left[f(x,y)\right]$$

$$s = T\left[r\right]$$

neighborhood size 1×1, *i.e.* a single pixel



Linear
Identity and negative

Logarithmic
Dynamic-range compression

Power-law
Adjustable exponent

Local intensity differences depend on the slope of T.

Image Negatives

- Image with intensity levels r in the range [0, L-1]
- Intensity levels of the output image: s = L-1-r
- Enhance white or gray detail embedded in dark regions of an image

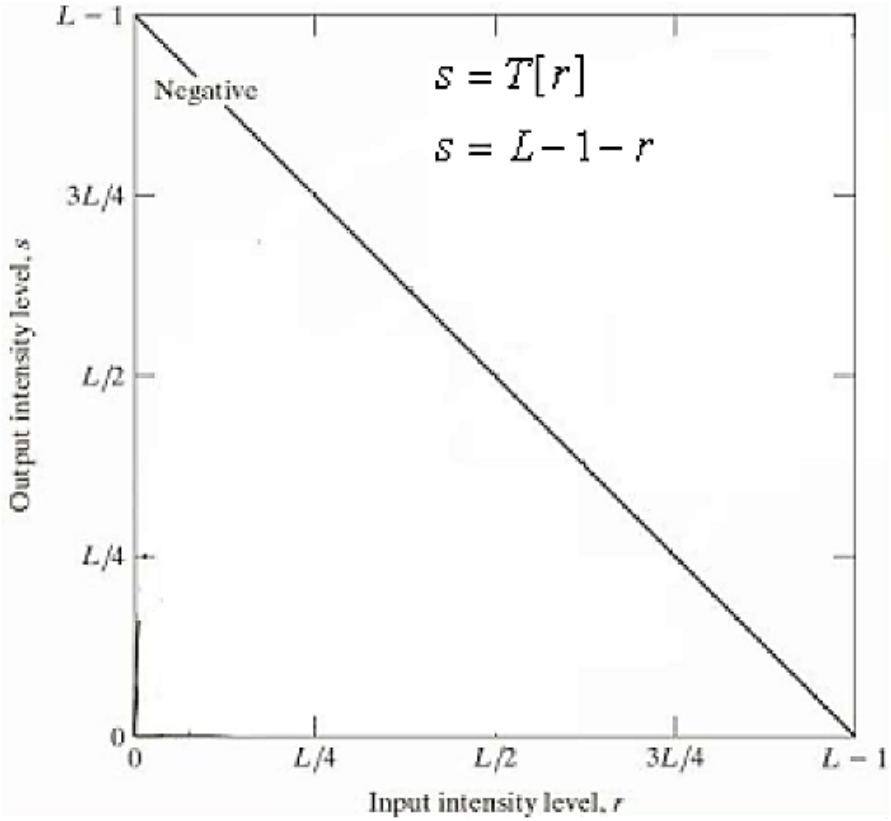
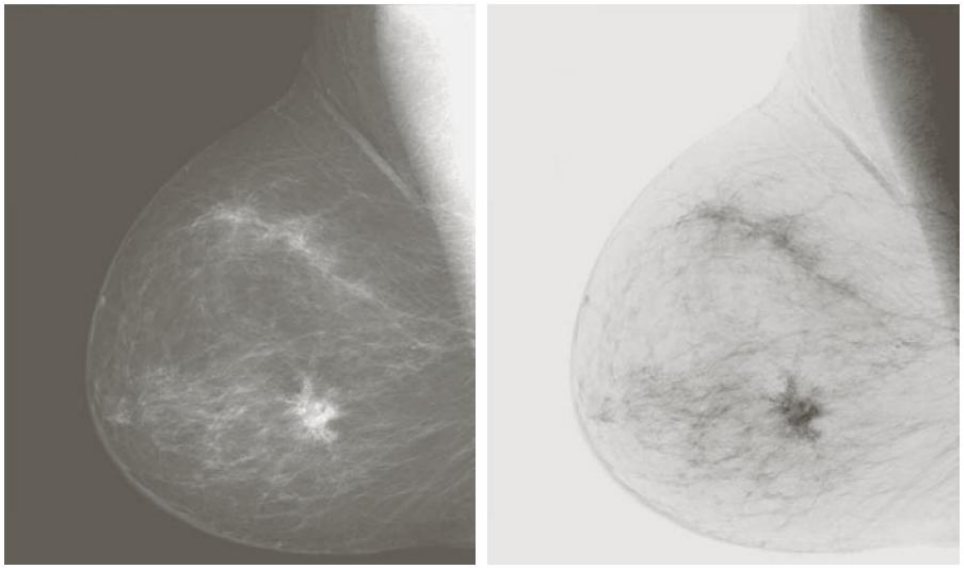


Image Negatives: Examples



Note how much clearer the tissue is in the negative image of the mammogram.

"*It makes all the difference whether one see darkness through the light or brightness through the shadows.*" David Lindsay



Log Transformations

- Log Transformation

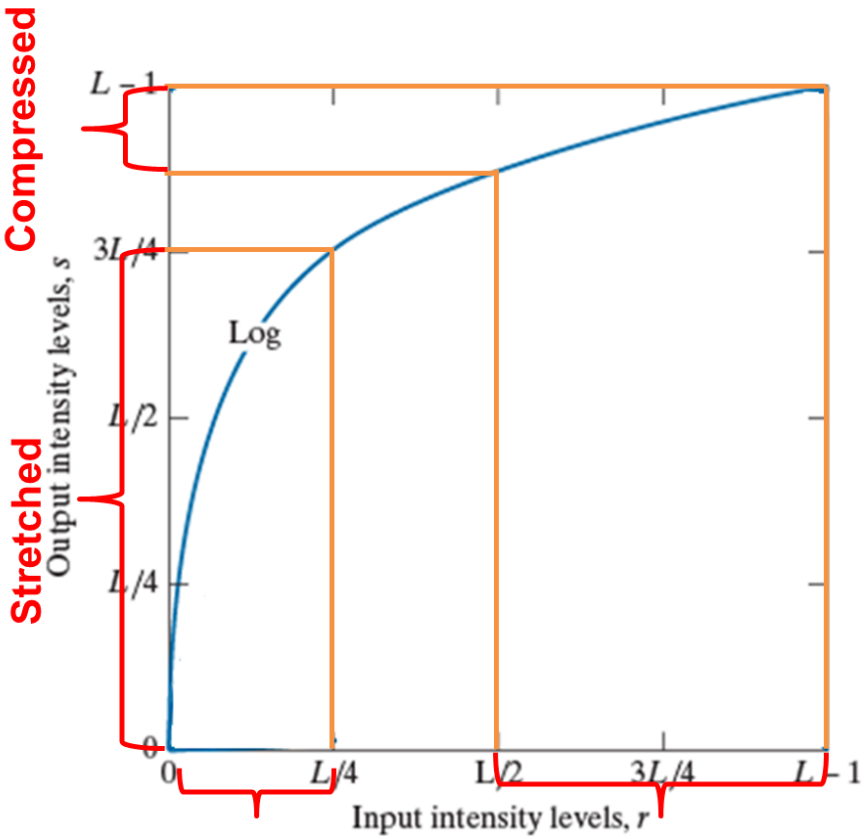
$$s = clog(1 + r)$$

- Maps a narrow range of low intensity values in the input into a wider range of output levels

Enhance low intensity values

- Maps a wide range of high intensity values in the input into a narrower range of out levels

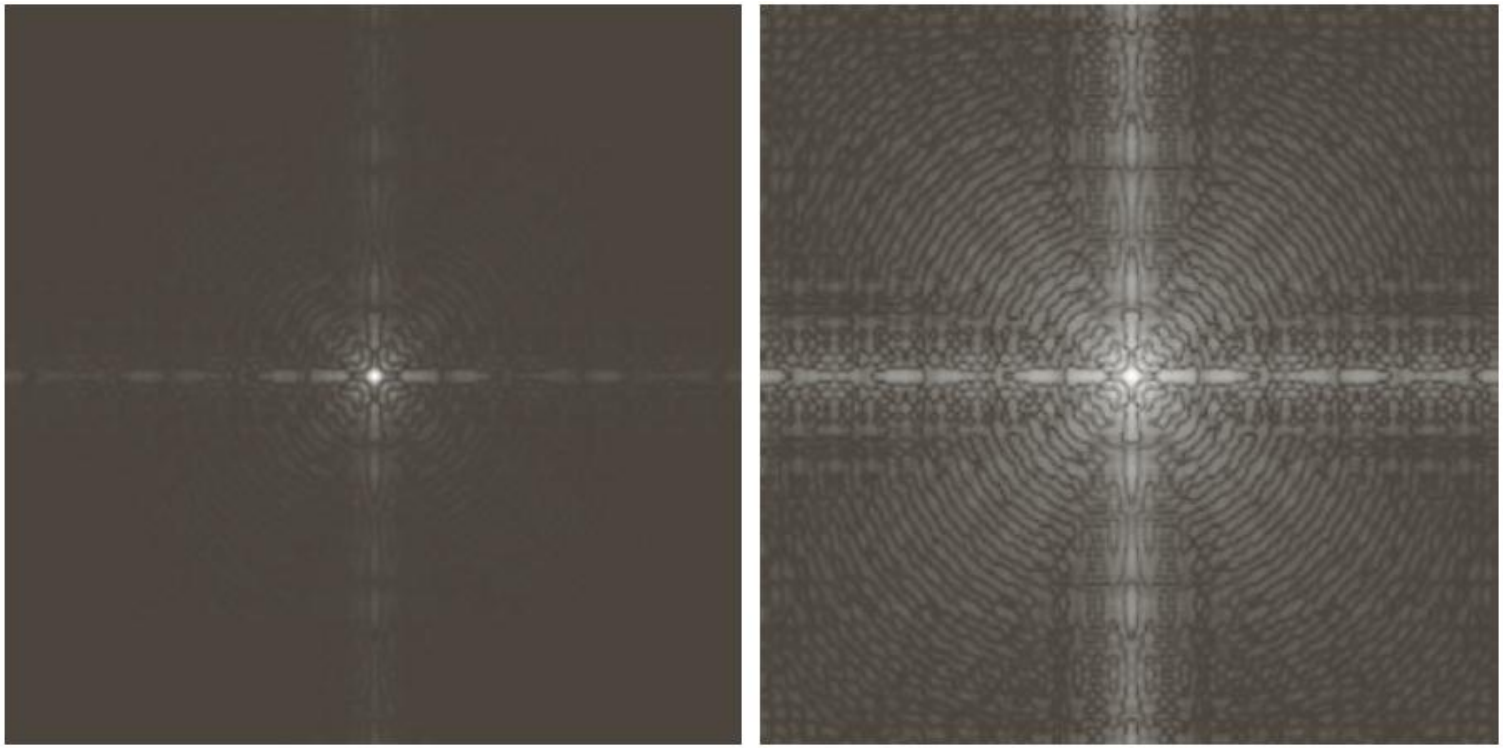
Suppress high intensity values



Logarithmic Spectrum Display

a b

FIGURE 3.5
(a) Fourier spectrum displayed as a grayscale image. (b) Result of applying the log transformation in Eq. (3-4) with $c = 1$. Both images are scaled to the range [0, 255].



$0 - 10^6$

Applications: Compress the dynamic range of an image

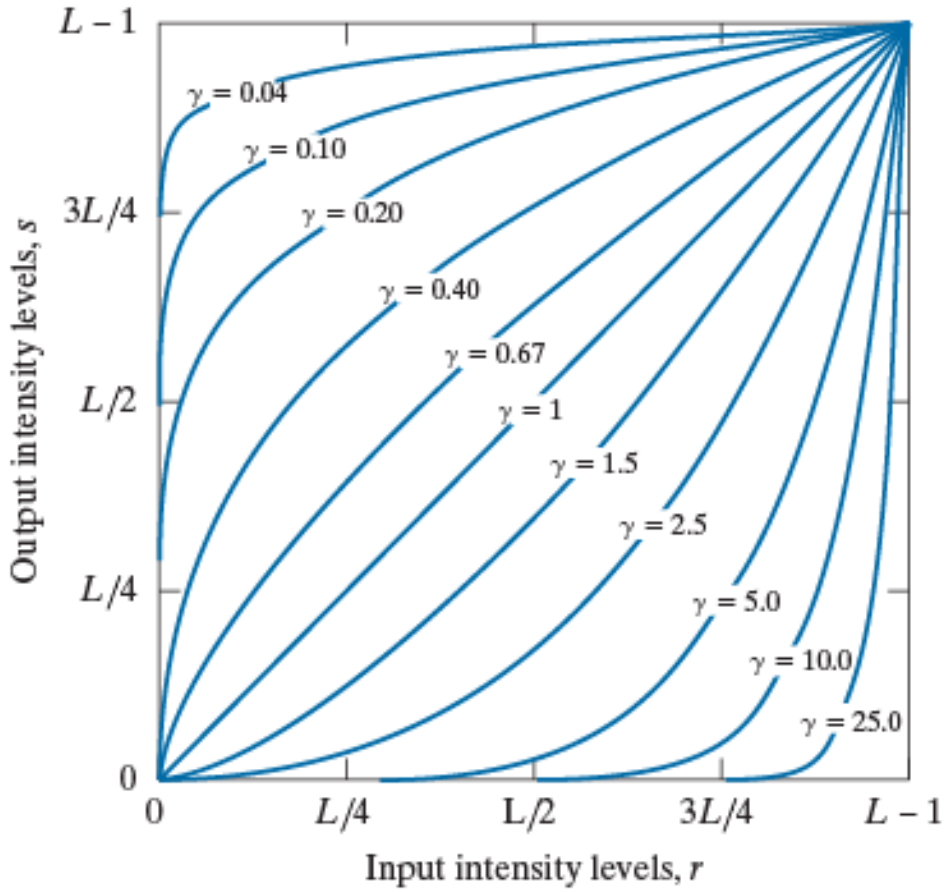
Power-Law Transformations

- Gamma Transformation

$$s = cr^{\gamma}$$

- $\gamma < 1$: Enhance low intensity values

- $\gamma > 1$: Enhance high intensity values



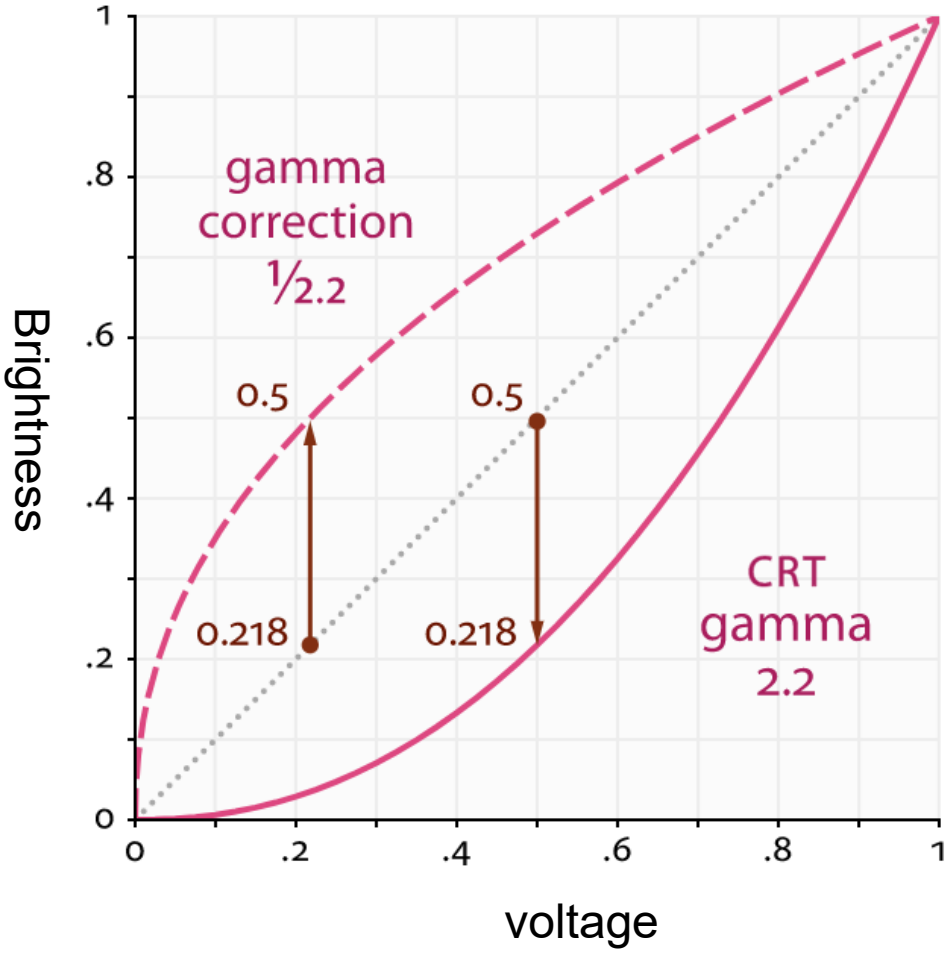
Gamma Correction

- CRT(阴极射线管) devices have an intensity to voltage response that is a power function:

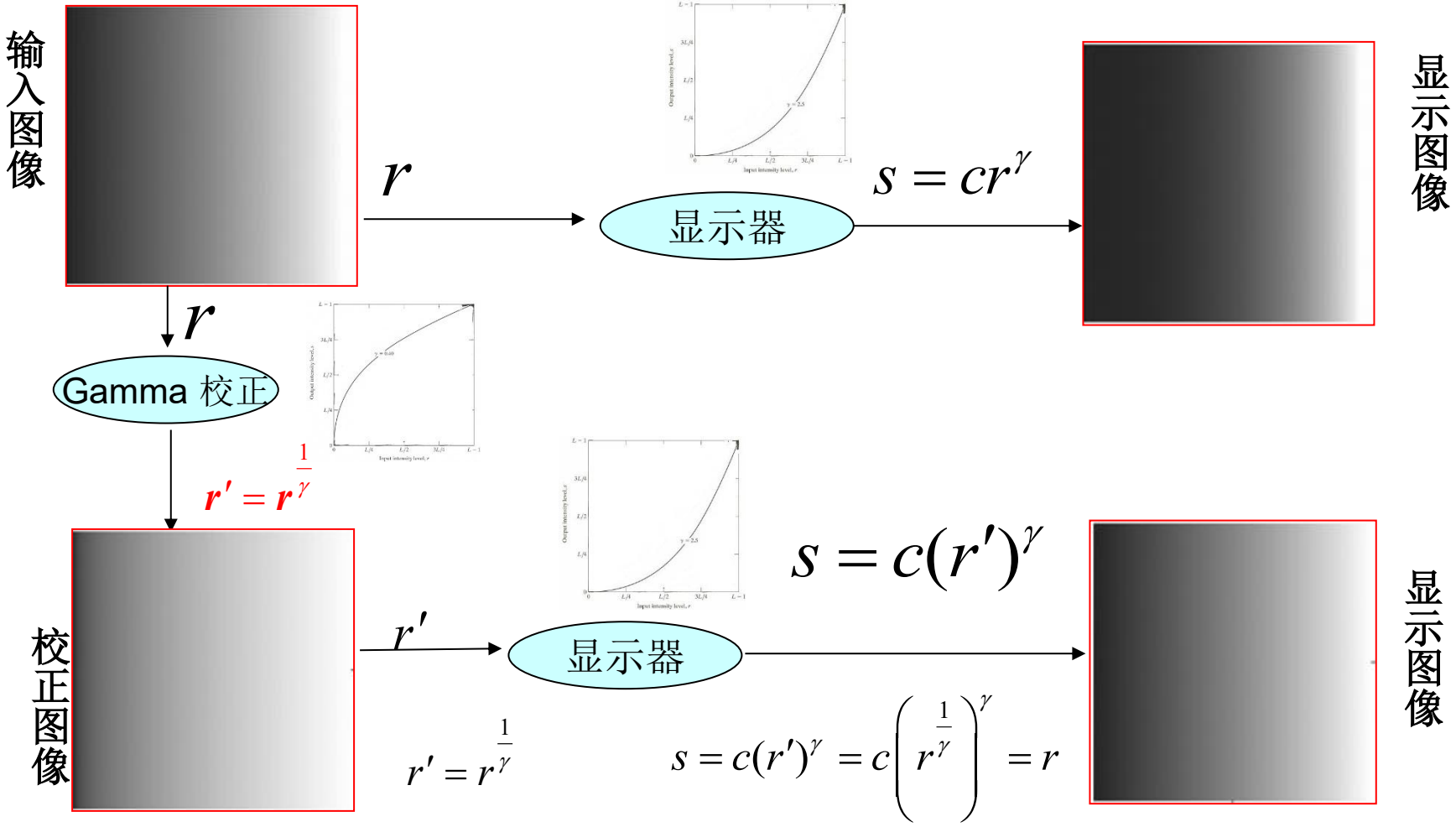
  $$s = r^{\gamma}, \gamma = 1.8 \sim 2.5$$

- Tend to produce images that are darker than intended, the lower intensity values are compressed

- Preprocess the input image before inputting it into the monitor: $s = r^{1/2.5} = r^{1/2.5}$

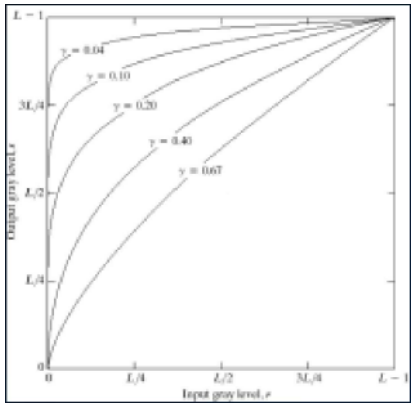


Gamma Correction

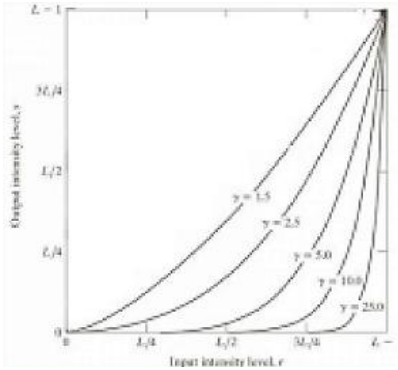


Gamma: Dark-Region Detail

- Inspect dark-region detail.



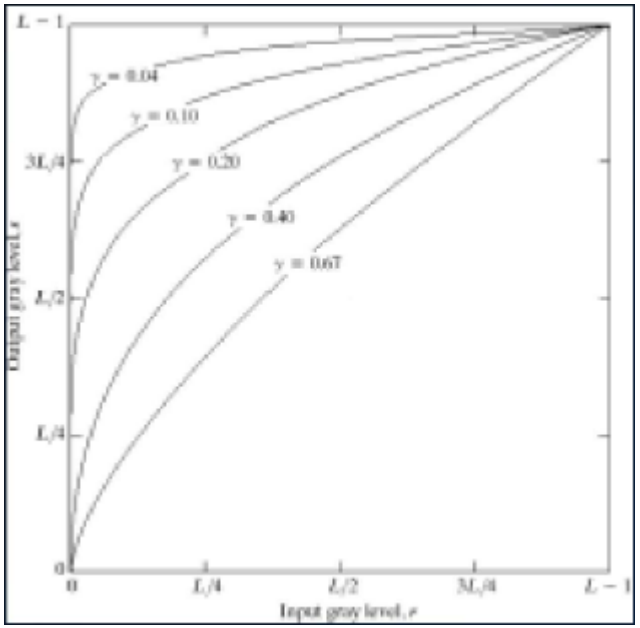
or



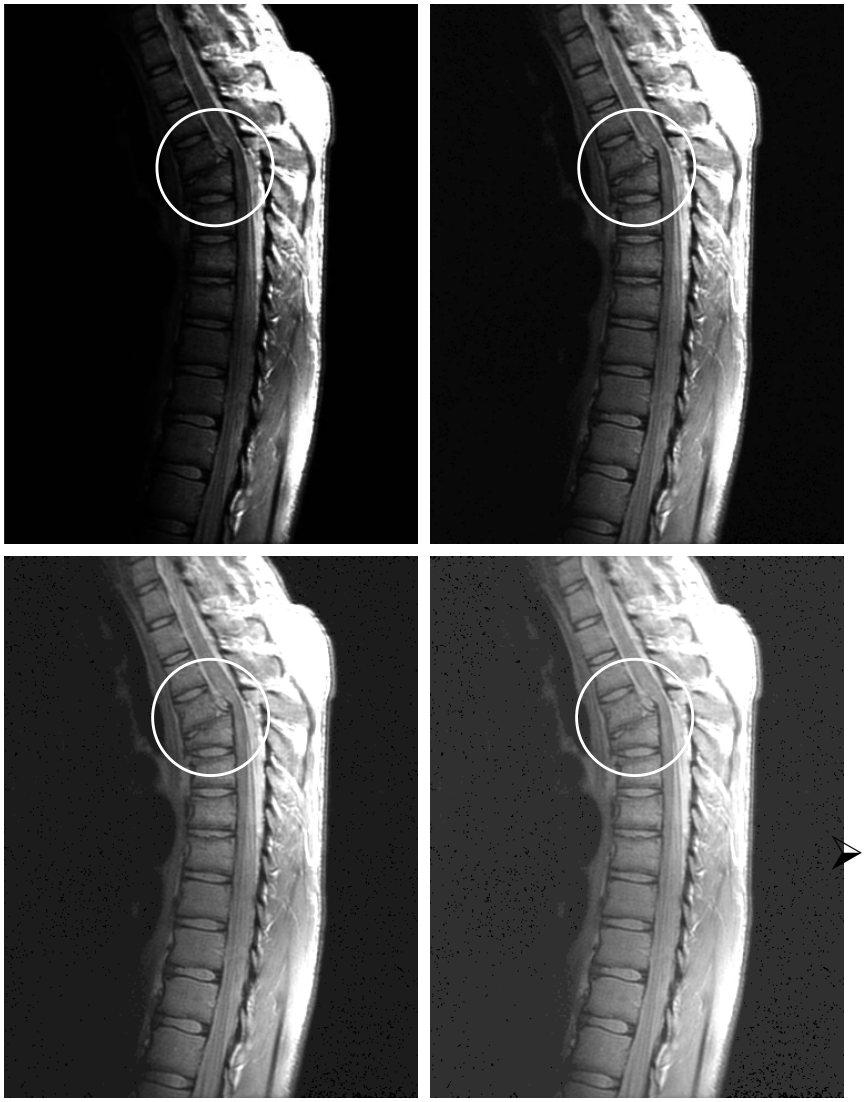


Gamma: Comparing Parameters

- Inspect dark-region detail.



$\gamma = 0.4$

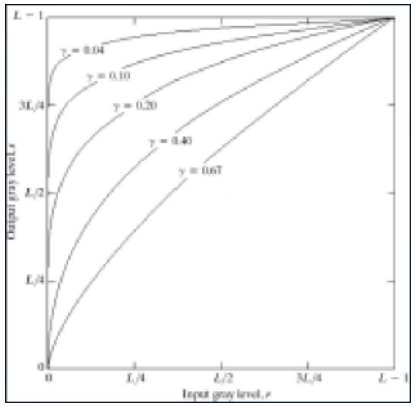


$\gamma = 0.6$

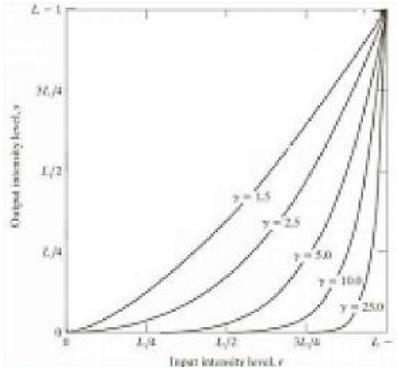
$\gamma = 0.3$

Gamma: Bright-Region Detail

- Inspect bright-region detail.



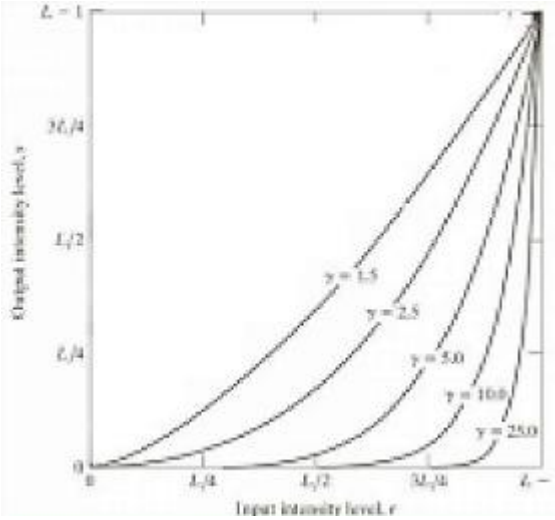
or





Gamma: Comparing Bright Regions

- Inspect bright-region detail.



$\gamma = 4$



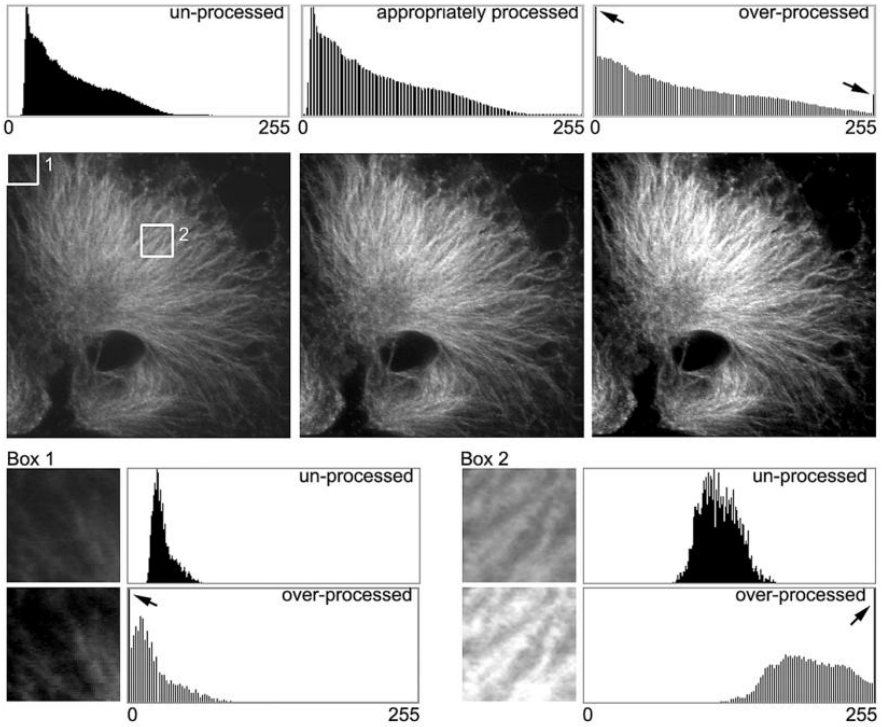
$\gamma = 3$

$\gamma = 5$

Scientific Image Processing

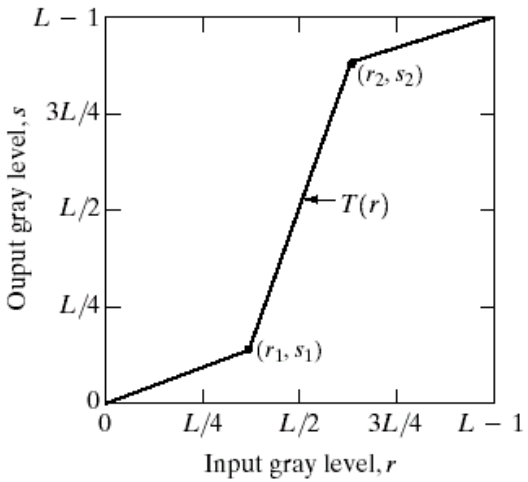
Avoiding Twisted Pixels: Ethical Guidelines for the Appropriate Use and Manipulation of Scientific Digital Images

Douglas W. Cromey



Piecewise-Linear Transformations

- Divide the intensity levels into different ranges, and use different linear transformation in different ranges
  - Pros: The form of piecewise functions can be arbitrarily complex
  - Cons: Considerably more user input

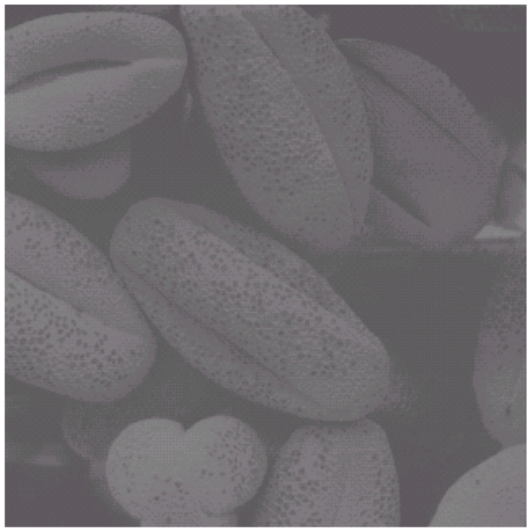


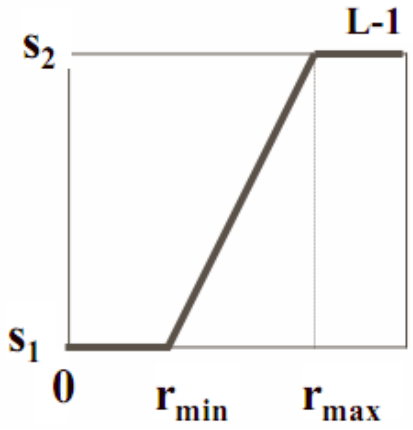
- Applications
  - Contrast stretching
  - Intensity-level slicing
  - Bit-plane slicing

$$s = T(r) = \begin{cases} a_1 r & 0 \le r < r_1 \\ a_2(r - r_1) + s_1 & r_1 \le r < r_2 \\ a_3(r - r_2) + s_2 & r_2 \le r \le L-1 \end{cases}$$

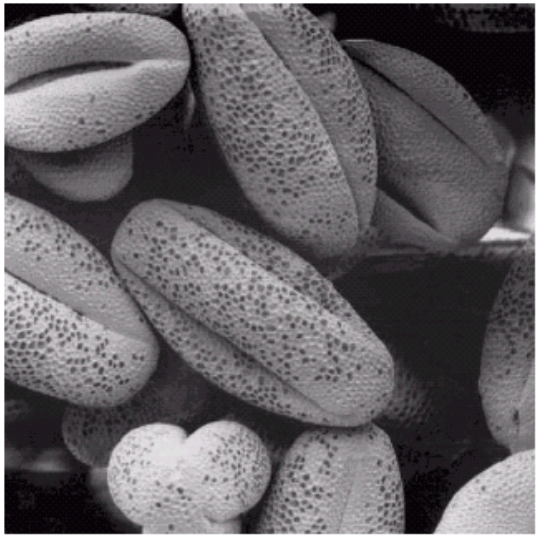
Stretching or Thresholding?

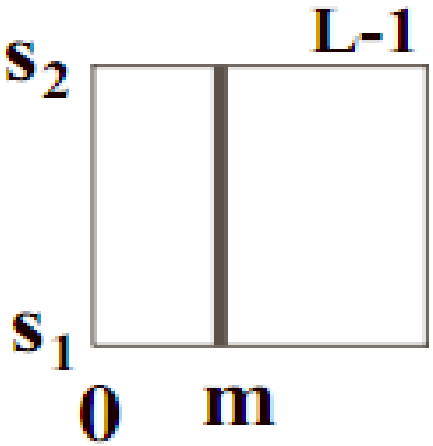
- Expand the range of intensity levels in an image









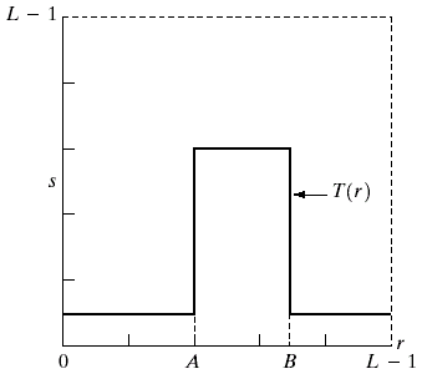


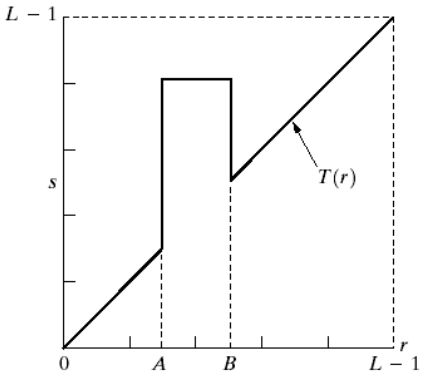


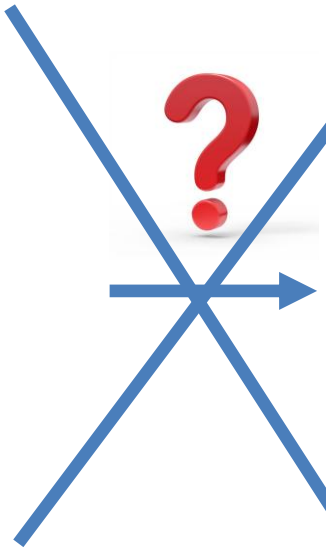
Intensity-Level Slicing

- Highlighting a specific range of intensities in an image that is of interest.







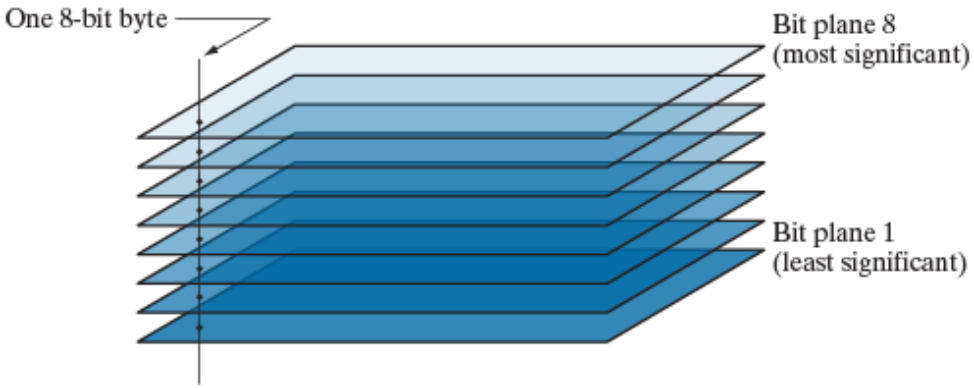


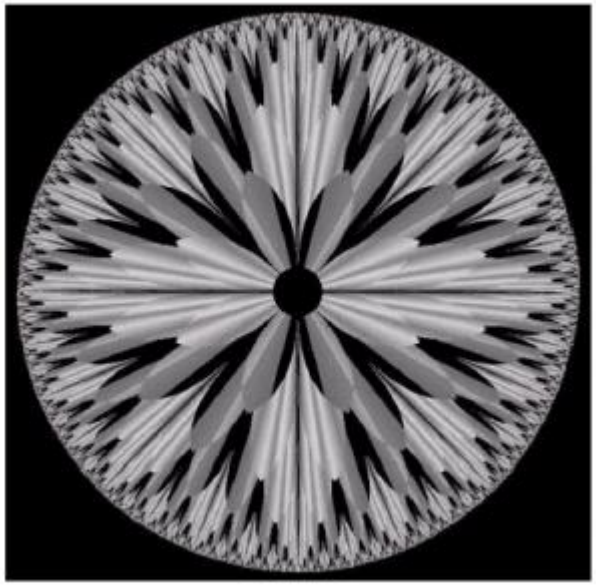




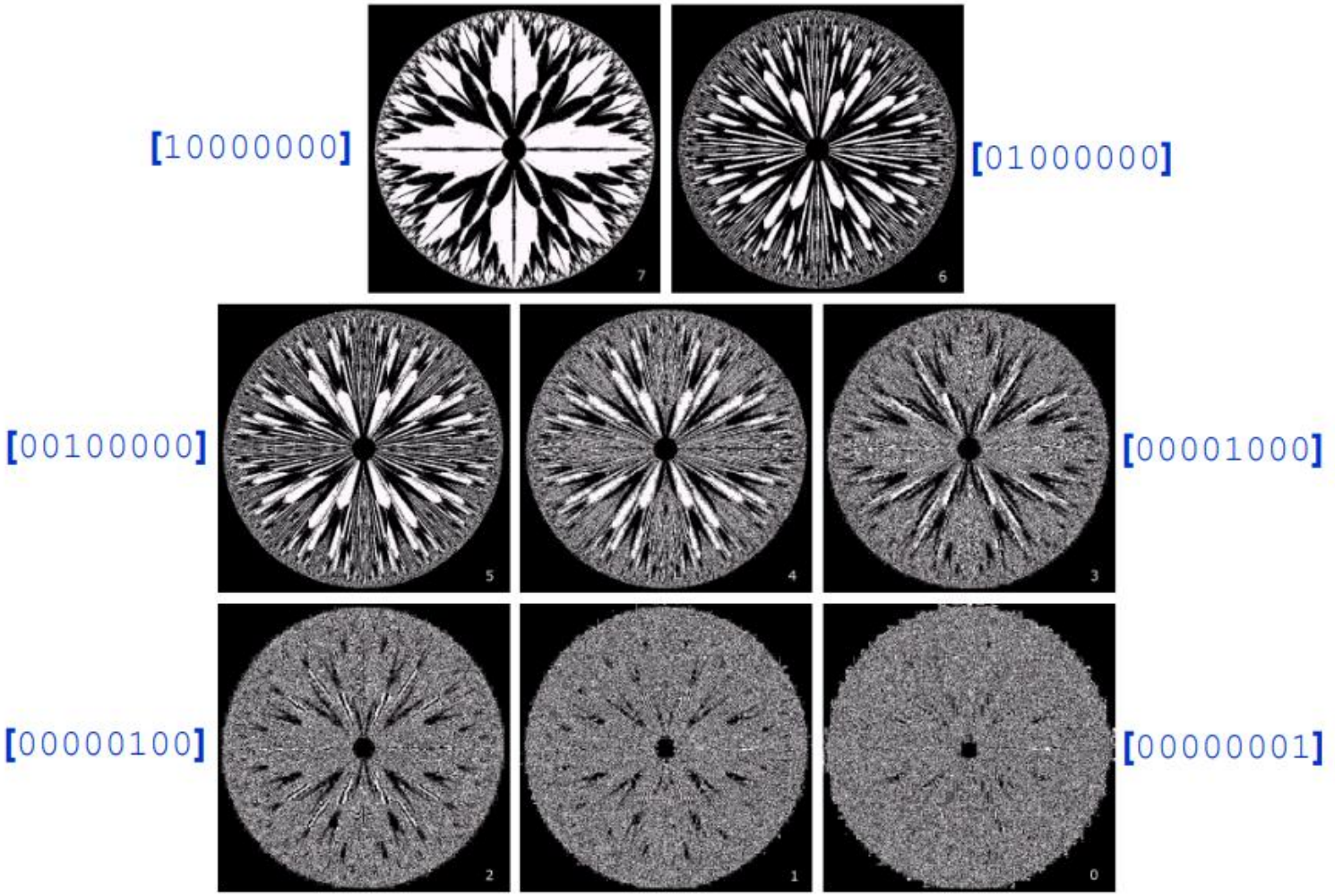
Bit-Plane Slicing

- Often by isolating particular bits of the pixel values in an image we can highlight interesting aspects of that image
  - Higher-order bits often show the main image structure
  - Lower-order bits may contain details and noise

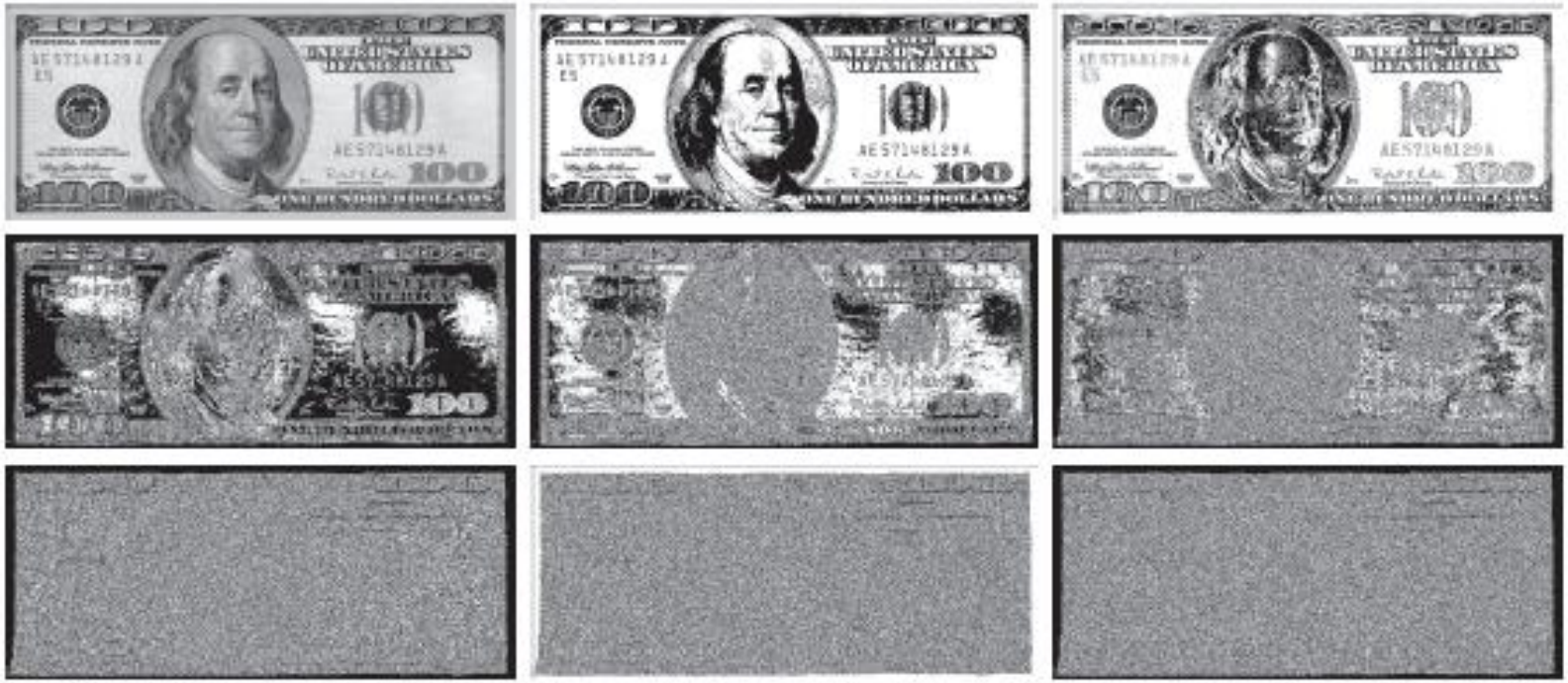




Bit Plane Slicing



Bit Planes: An Example



a b c
d e f
g h i

FIGURE 3.14 (a) An 8-bit gray-scale image of size 550×1192 pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image..

Reconstruction from Bit Planes



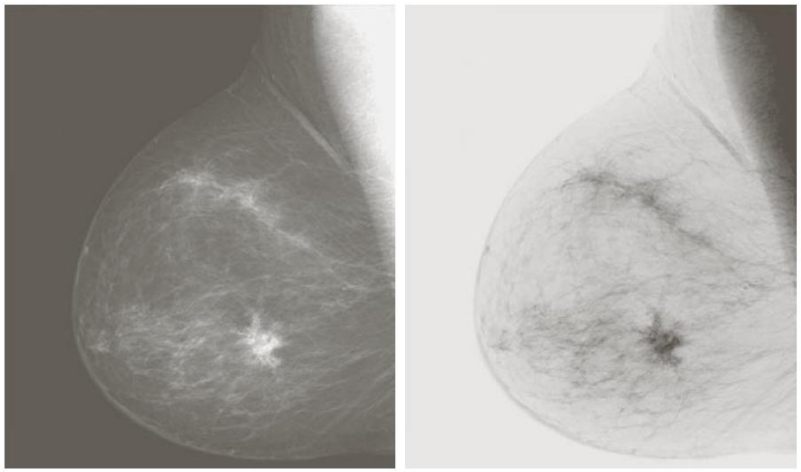
a b c **FIGURE 3.15** Image reconstructed from bit planes: (a) 8 and 7; (b) 8, 7, and 6; (c) 8, 7, 6, and 5.

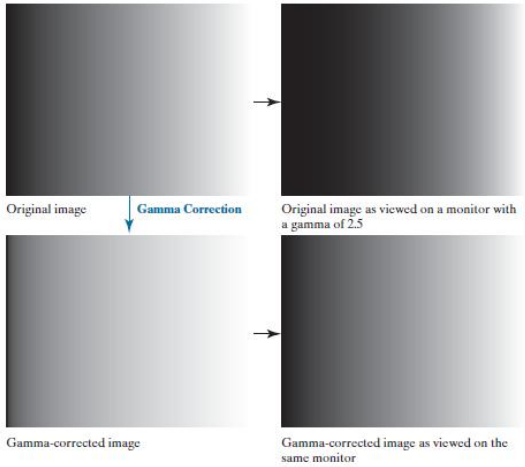
Useful for compression.

Reconstruction is obtained by:

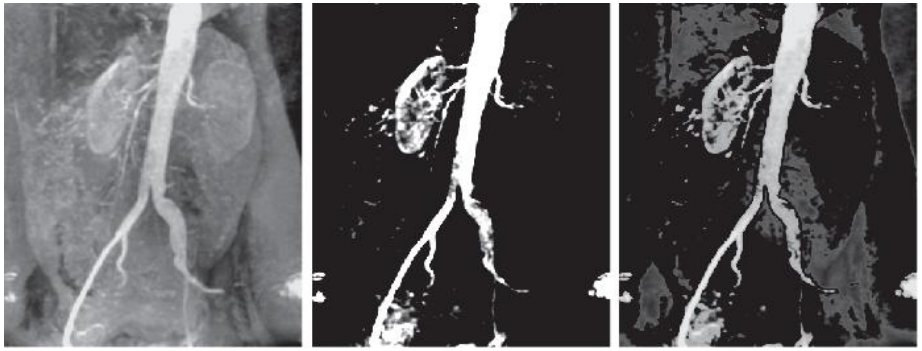
$$I(i,j) = \sum_{n=1}^{N} 2^{n-1} I_n(i,j)$$

Summary of Basic Intensity Transformation











- What are changed by intensity transform
- What's the drawback? (not automatic)



From Chosen to Estimated Mappings

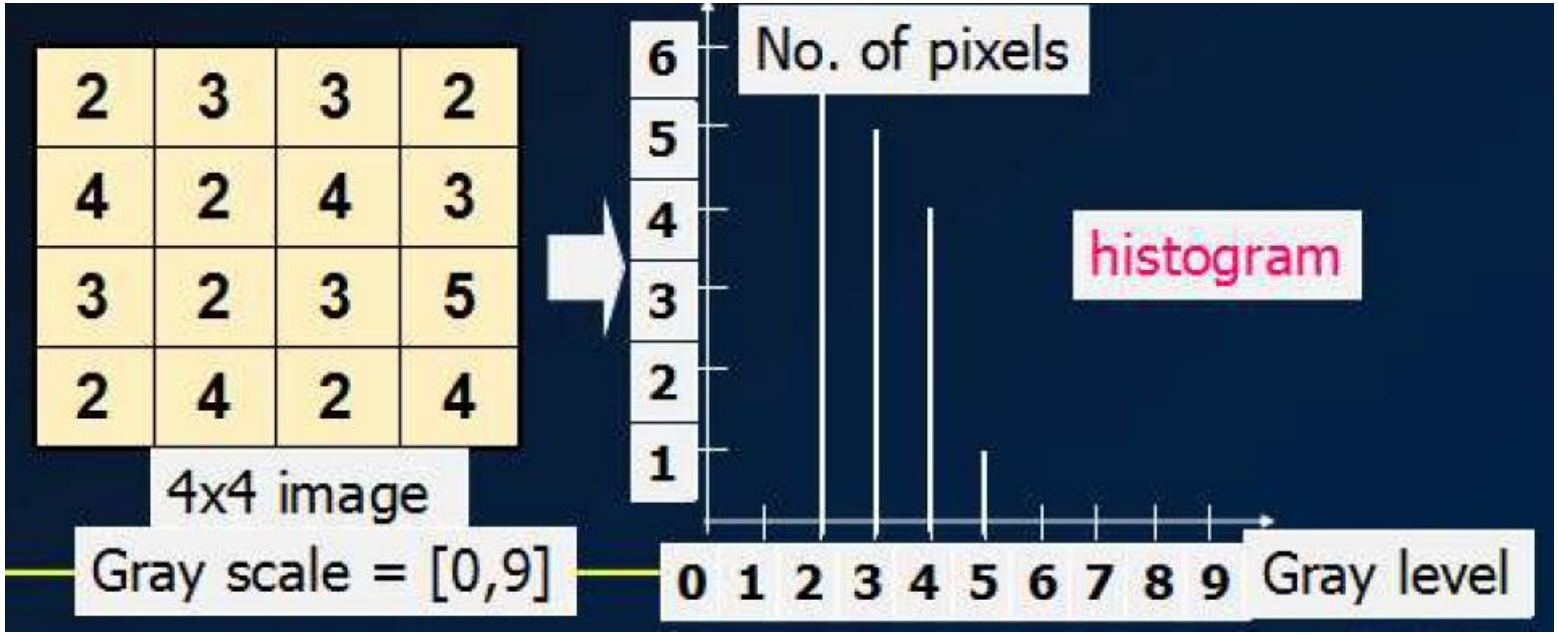
- So far, we have chosen a mapping and its parameters.
- Image statistics can help estimate a mapping automatically.
- The histogram records how often each intensity occurs.

Can the intensity distribution determine T?

Histogram Processing

Definition of histogram (hist = bar; gram = 图)

If the intensity levels is in the rang [0,L-1], the histogram is a discrete function $\mathrm{h}(r_k) = n_k$, $(k = 0,1,\ldots,L-1)$, where r_k is the kth intensity value, and n_k is the number of pixels in the image with intensity r_k. A normalized histogram is given by $\mathrm{p}(r_k) = \frac{n_k}{MN}$, where M and N are the row and column dimensions of the image.



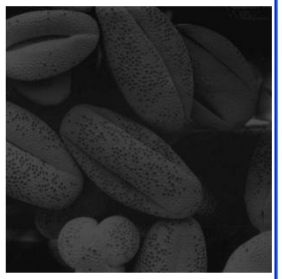
Histogram Processing

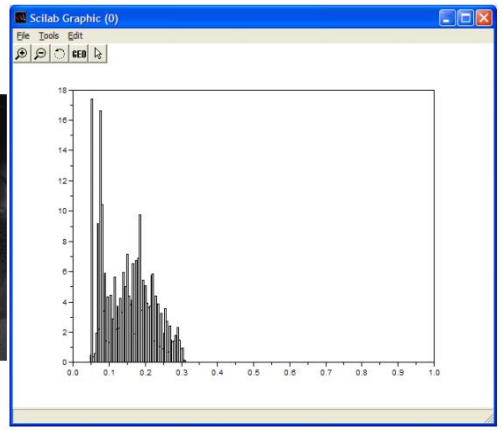
Motivation

Intensity transform can enhance image because it properly changes the image histogram. So we can directly design an intensity transform function based on histogram

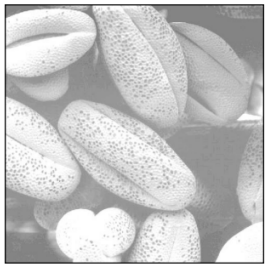
Histogram Examples

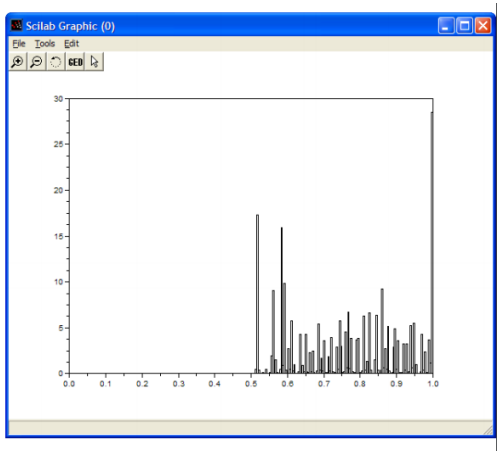
Dark image



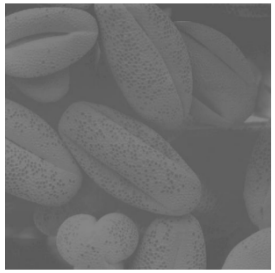


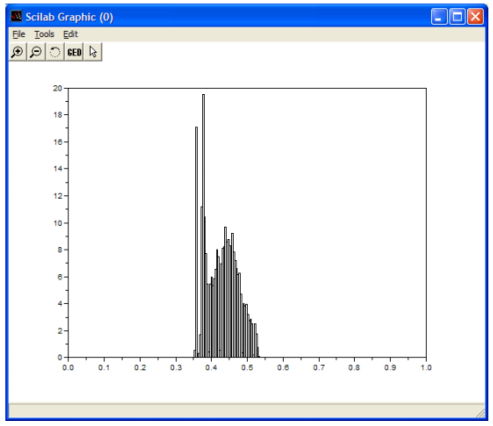
Bright image





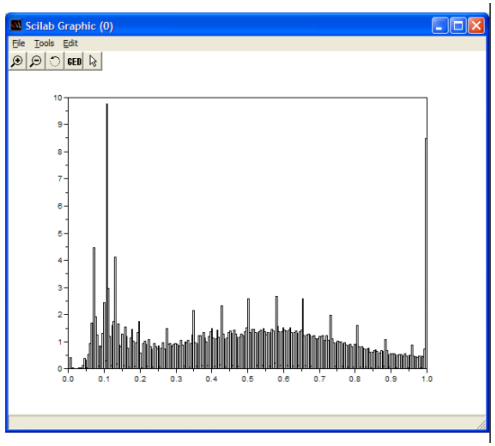
Low contrast





Broad intensity range





Same Histogram, Different Images?

Image A

0	1	0	1
1	0	1	0
0	1	0	1
1	0	1	0

Image B

0	0	1	1
0	0	1	1
0	0	1	1
0	0	1	1

Do the histograms match? Do the structures match?
Would global histogram equalization use the same mapping?

What the Histogram Omits

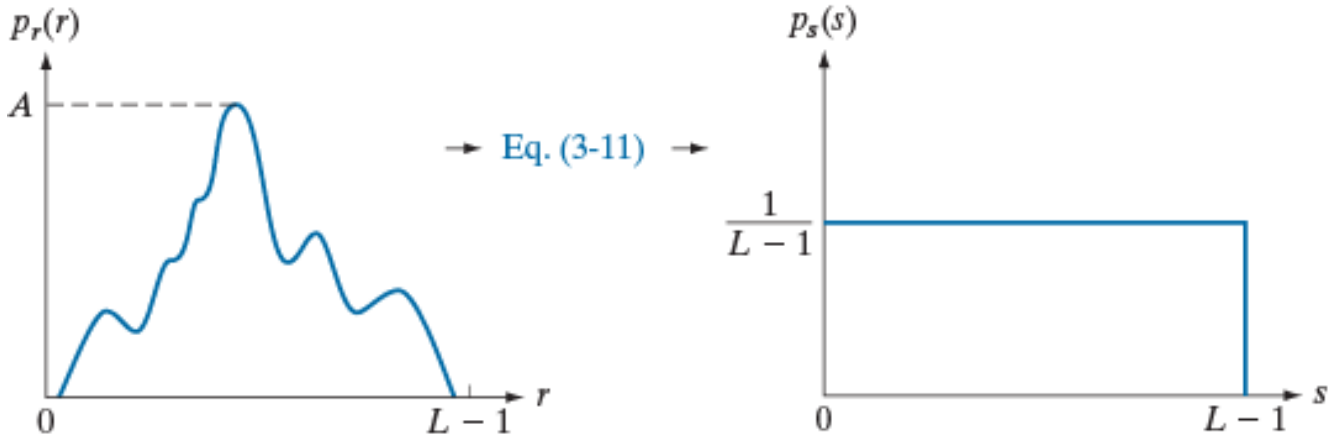
Quantity	Image A	Image B
Number of black pixels	8	8
Number of white pixels	8	8
p(0), p(1)	0.5, 0.5	0.5, 0.5
Spatial arrangement	Alternating	Two regions

A histogram discards pixel locations.
Matching histograms does not imply matching structures.

Histogram Equalization (HE)

- What is histogram equalization

A process that seek an intensity transform function $s = T(r)$ so that the histogram of transformed image becomes flat.



a b

FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying Eq. (3-11) to the input PDF. The resulting PDF is always uniform, independently of the shape of the input.

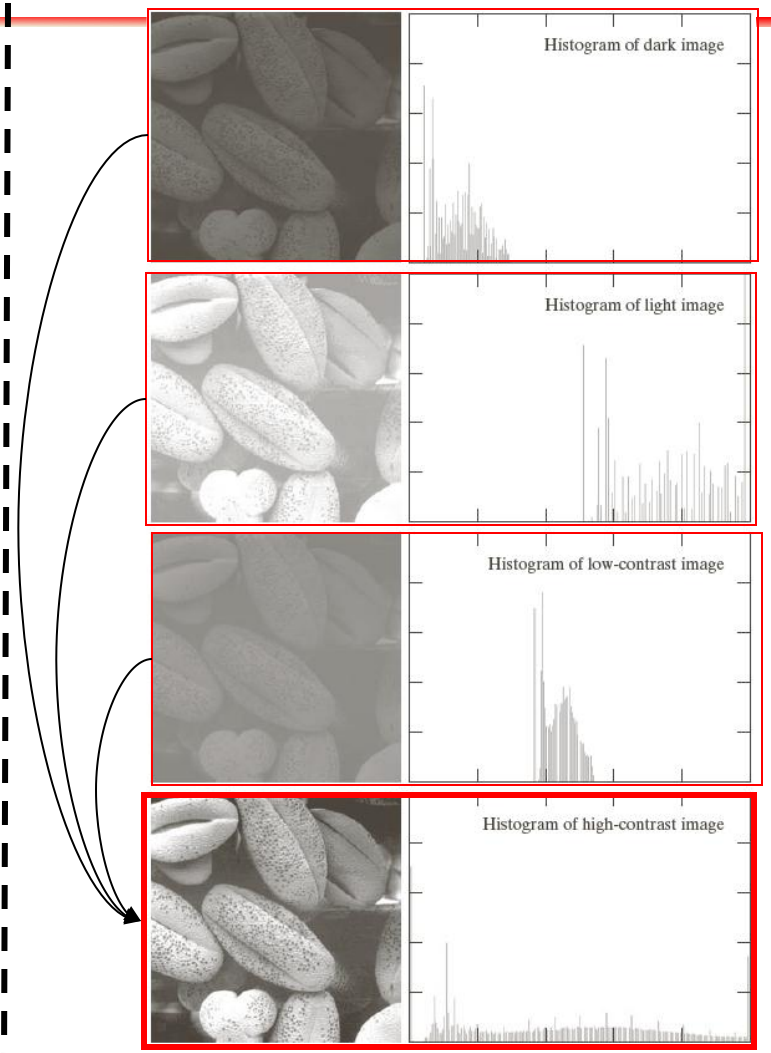
Histogram Equalization (HE)



The advantages of HE

Automatically determine an optimal transform function

Spreading out the frequencies in an image (or equalizing the image) is a simple way to improve dark or washed out images.



Histogram Equalization (HE)

- At first, the continuous case will be studied:
  - r is the intensity of the image in $[0, L-1]$
  - We focus on transformation $s = T(r)$:
- $T(r)$ is strictly monotonically increasing.
- $T(r)$ must satisfy:

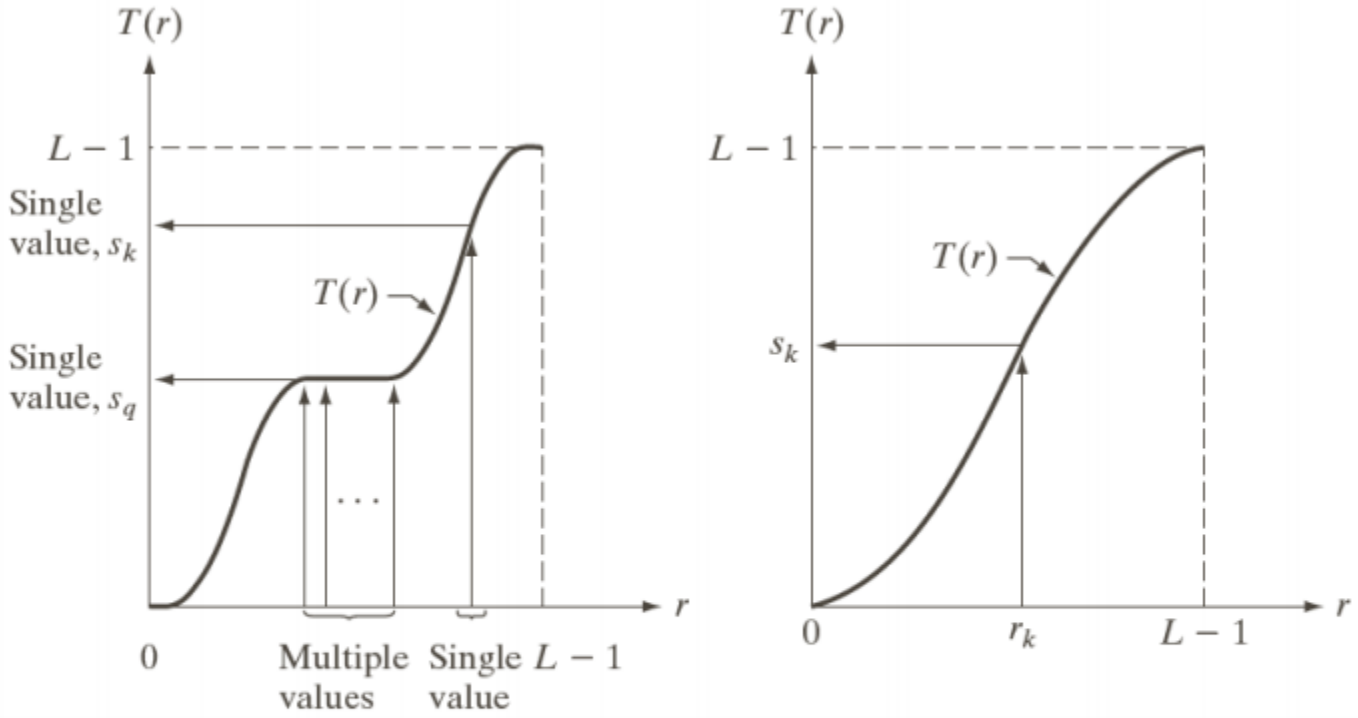
$$0 \le T(r) \le L-1, \ for \ \ 0 \le r \le L-1$$

Histogram Equalization (HE)

The constraints of HE

- The condition for $T(r)$ to be monotonically increasing guarantees that ordering of the output intensity values will follow the ordering of the input intensity values (avoids reversal of intensities).
- If $T(r)$ is strictly monotonically increasing then the mapping from s back to r will be 1-1.
- The second condition ($T(r)$ in $[0,1]$) guarantees that the range of the output will be the same as the range of the input.

Histogram Equalization (HE)



a) We cannot perform inverse mapping (from s to r).

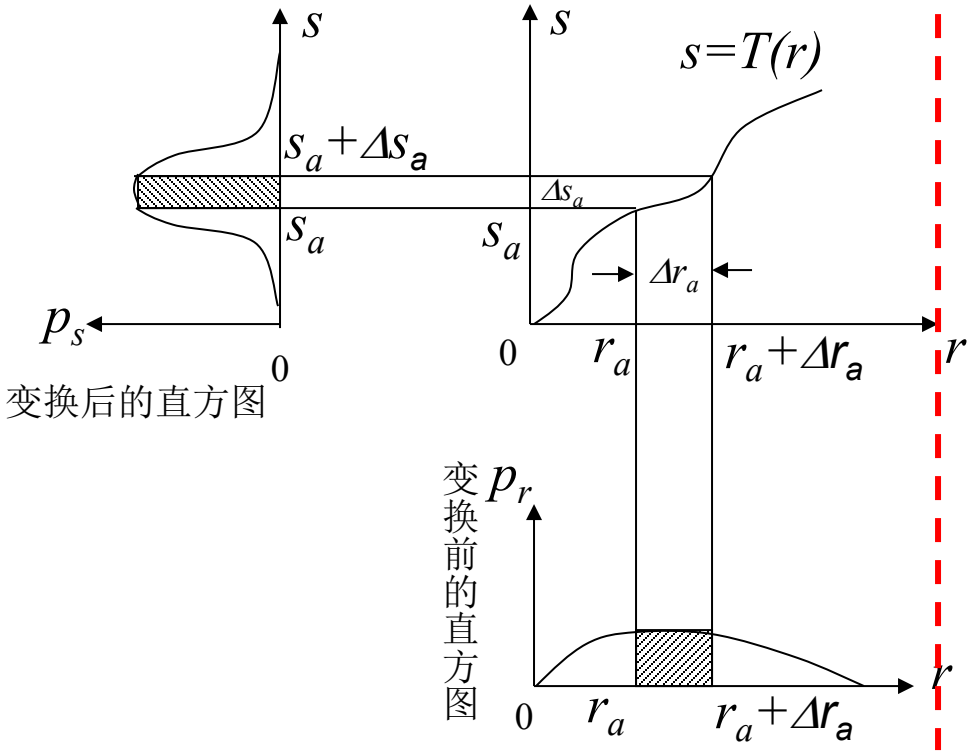
b) Inverse mapping is possible.

Histogram Equalization (HE)

➢**What's the relationship of the original histogram and the transformed one?**

用r表示变换前的灰度，$p_r(w)$ 表示变换前的直方图

用s表示变换后的灰度，$p_s(w)$ 表示变换后的直方图



变换前后面积相等

$$\int_{s_a}^{s_a+\Delta s_a} p_s(w)dw = \int_{r_a}^{r_a+\Delta r_a} p_r(w)dw$$

如果Δr_a小，Δs_a也小

$$p_s(s_a)\Delta s_a = p_r(r_a)\Delta r_a$$

$$p_s(s_a) = \frac{p_r(r_a)}{\Delta s_a / \Delta r_a} = \frac{p_r(r_a)}{ds_a / dr_a}$$

其中 $\frac{ds_a}{dr_a} = \frac{dT(r_a)}{dr_a}$

故：$p_s(s_a) = p_r(r_a)\frac{1}{dT(r_a)/dr_a}$ 

变换后s的PDF由变换前r的ＰＤＦ和所用的变换函数Ｔ决定。

Histogram Equalization (HE)

How to HE based on the relationship $p_s(s) = p_r(r)\dfrac{1}{dT(r)/dr}$

The formula of HE

If $s = T(r) = (L-1)\int_0^r p_{r(w)}dw$, then $p_s(s) = \dfrac{1}{L-1}$

Cumulative distribution function (CDF) is used as the HE transform function

Proof: Because $s = T(r) = (L-1)\int_0^r p_r(w)dw$

牛顿-莱布尼茨公式

so $$\frac{dT(r)}{dr} = \frac{d[(L-1)\int_0^r p_r(w)dw]}{dr} = (L-1)p_r(r)$$

substitute $p_s(s) = p_r(r)\dfrac{1}{dT(r)/dr}$ $= p_r(r) * \dfrac{1}{(L-1)p_r(r)} = \dfrac{1}{L-1}$

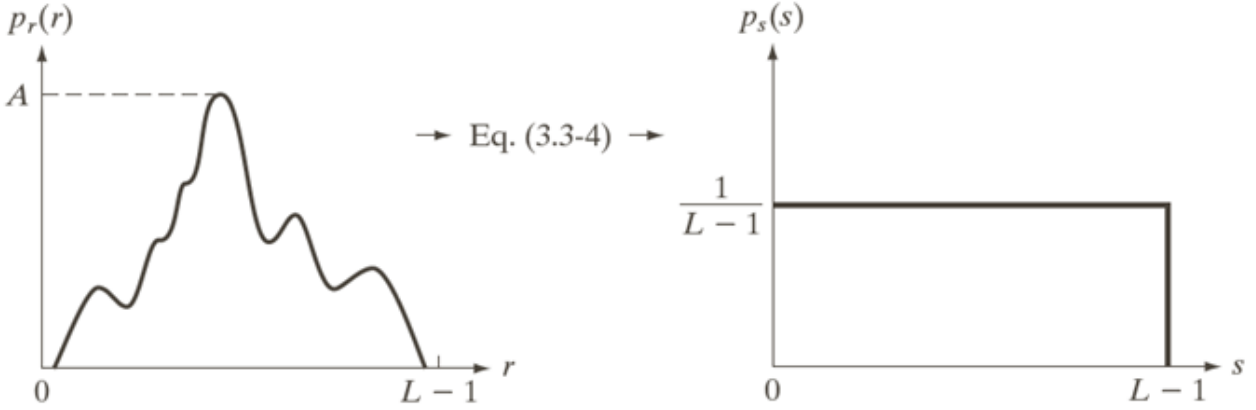
随机变量S具有均匀PDF表征

Histogram Equalization (HE)

If the formula of HE

If $s = T(r) = (L-1)\int_0^r p_{r(w)dw}$, then $p_s(s) = \frac{1}{L-1}$,0≤s ≤L-1

Cumulative distribution function (CDF) is used as the HE transform function



a b

FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying the transformation in Eq. (3.3-4) to all intensity levels, r. The resulting intensities, s, have a uniform PDF, independently of the form of the PDF of the r's.

Histogram Equalization (HE)

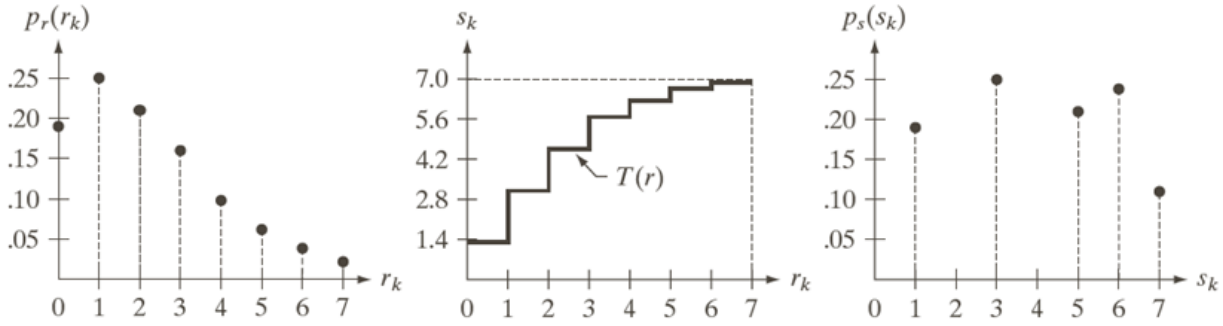
The formula of HE

If $s = T(r) = (L-1)\int_0^r p_{r(w)dw}$,then $p_s(s) = \frac{1}{L-1}$,0≤s ≤L-1

discrete

$$p_k(r_k) = \frac{n_k}{n} \quad k = 0,1,\ldots,L-1$$

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k} \frac{n_j}{n} \quad k = 0,1,\ldots,L-1$$



a b c

FIGURE 3.19 Illustration of histogram equalization of a 3-bit (8 intensity levels) image. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Histogram Equalization (HE)

The formula for histogram equalisation in the discrete case is given

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = \frac{(L-1)}{MN}\sum_{j=0}^{k} n_j$$

where

- r_k: input intensity
- s_k: processed intensity
- n_j: the frequency of intensity j
- MN: the number of image pixels.

Histogram Equalization (HE)

A 3-bit 64x64 image has the following intensities:

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j)$$

Applying histogram equalization:

$$s_0 = T(r_0) = 7\sum_{j=0}^{0} p_r(r_j) = 7\,p_r(r_0) = 1.33$$

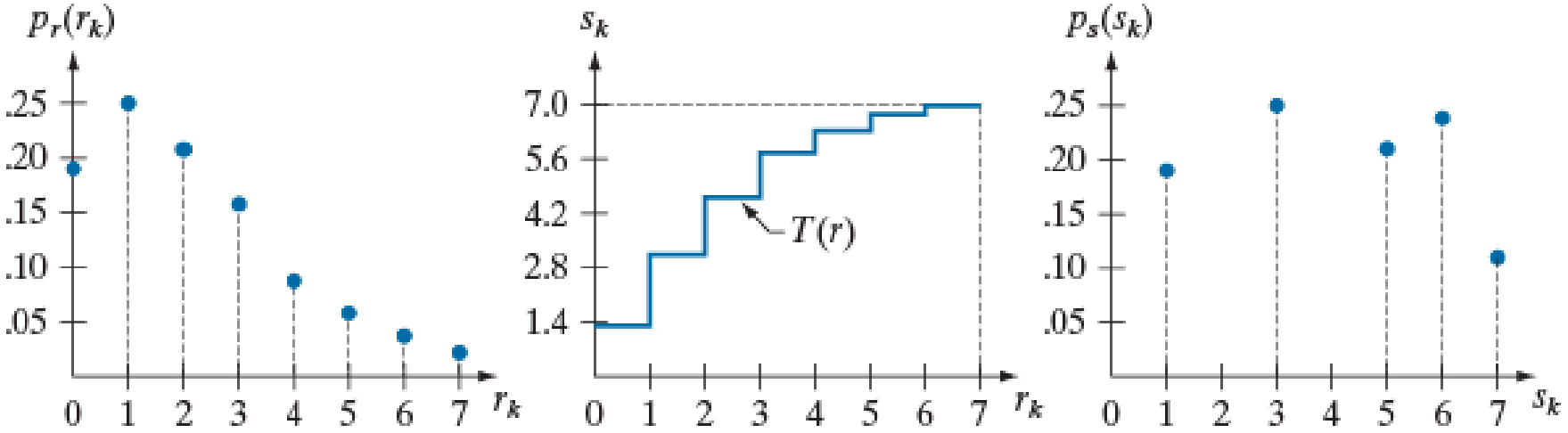
$$s_1 = T(r_1) = 7\sum_{j=0}^{1} p_r(r_j) = 7\,p_r(r_0) + 7\,p_r(r_1) = 3.08$$

Histogram Equalization (HE)

Rounding to the nearest integer:

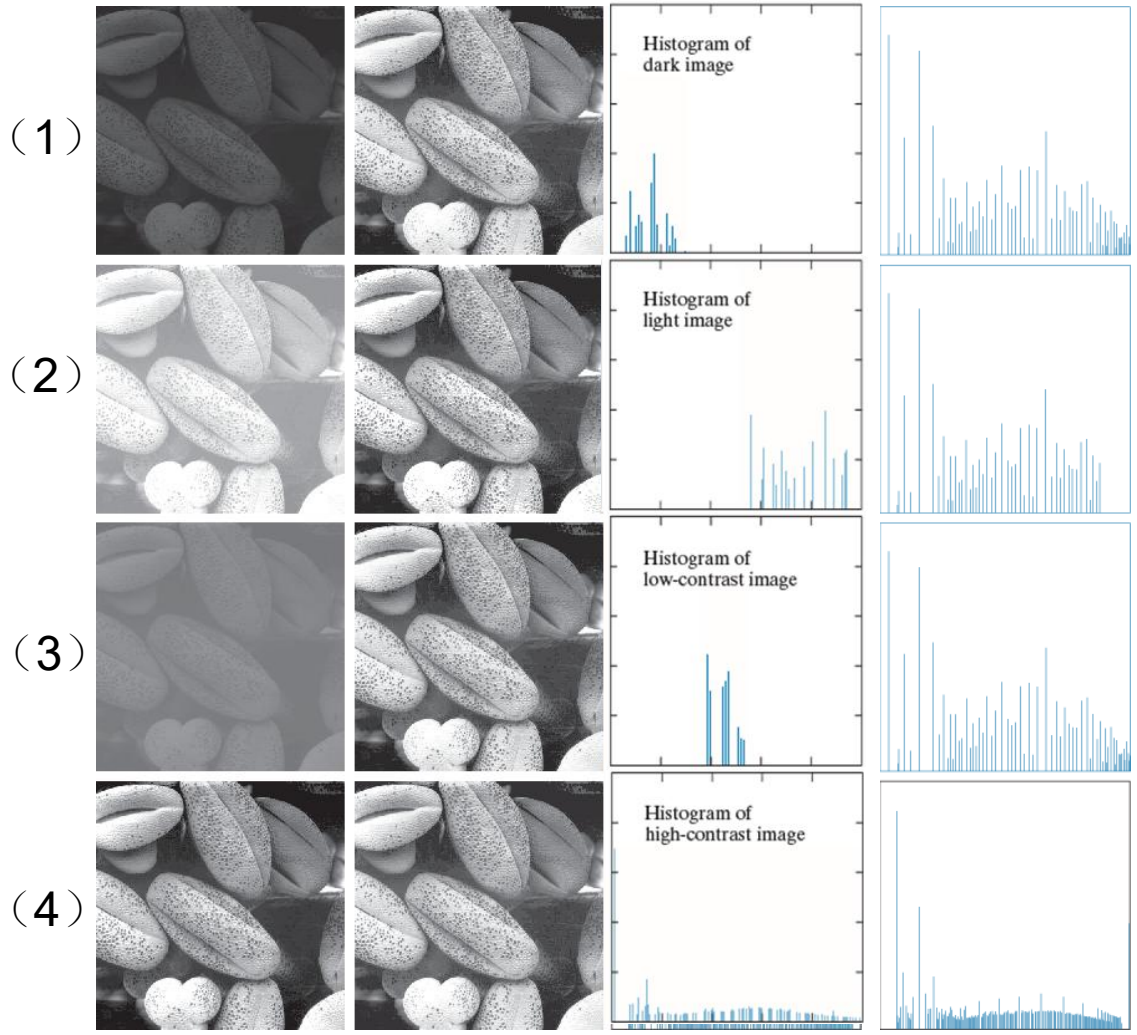
$$s_0 = 1.33 \rightarrow 1 \quad s_1 = 3.08 \rightarrow 3 \quad s_2 = 4.55 \rightarrow 5 \quad s_3 = 5.67 \rightarrow 6$$

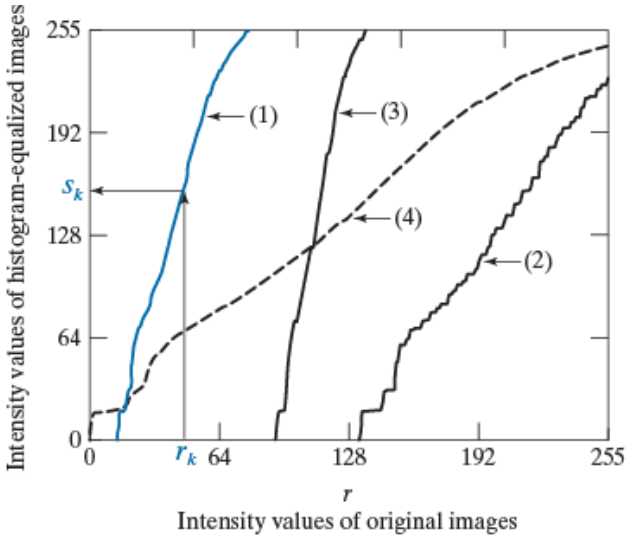
$$s_4 = 6.23 \rightarrow 6 \quad s_5 = 6.65 \rightarrow 7 \quad s_6 = 6.86 \rightarrow 7 \quad s_7 = 7.00 \rightarrow 7$$



Notice that due to discretization, the resulting histogram will rarely be perfectly flat. However, it will be extended.

Histogram Equalization (HE)

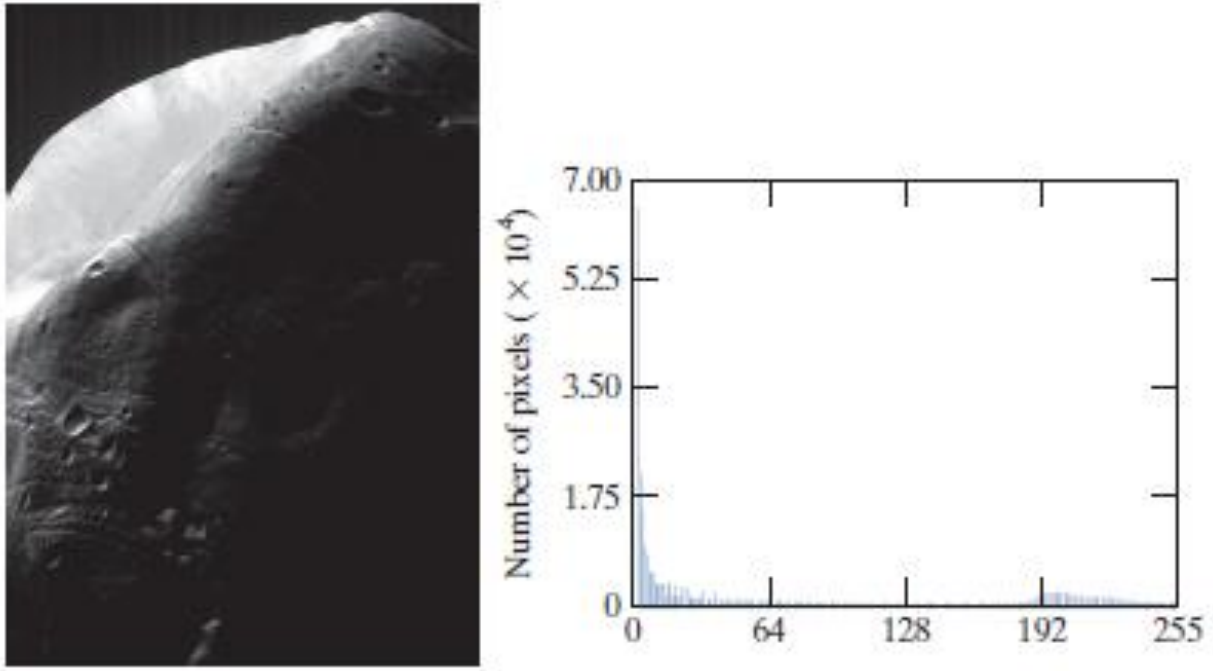




- Even though all these histograms are different, the histogram equalized images themselves are visually very similar
- Histogram equalization does not have much effect on images of high contrast

When Equalization Is Unsuitable

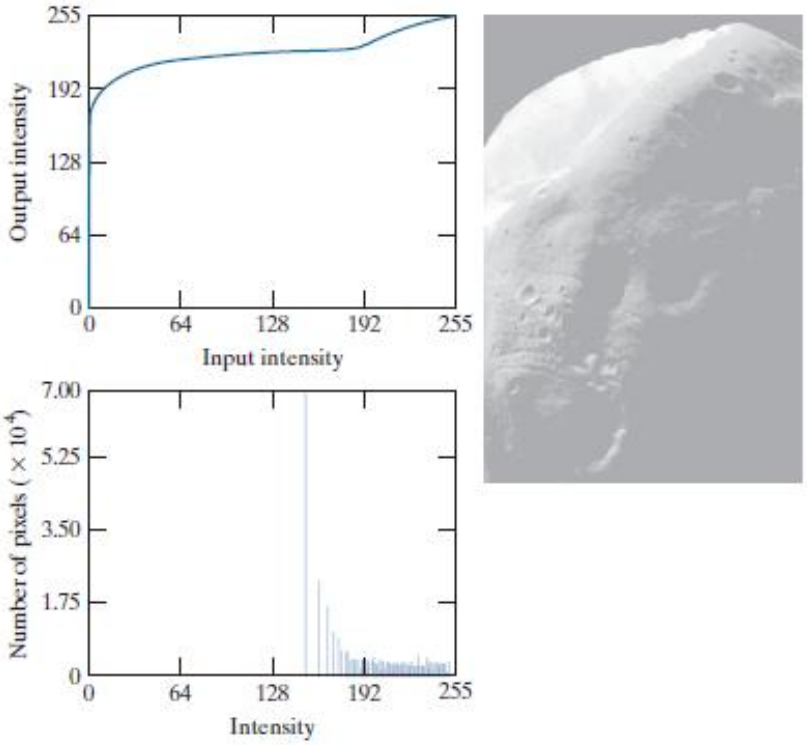
- Histogram equalization does not always provide the desirable results.



- Image of Phobos (Mars moon) and its histogram.
- Many values near zero in the initial histogram

HE: A Washed-Out Result

- Histogram equalization does not always provide the desirable results.

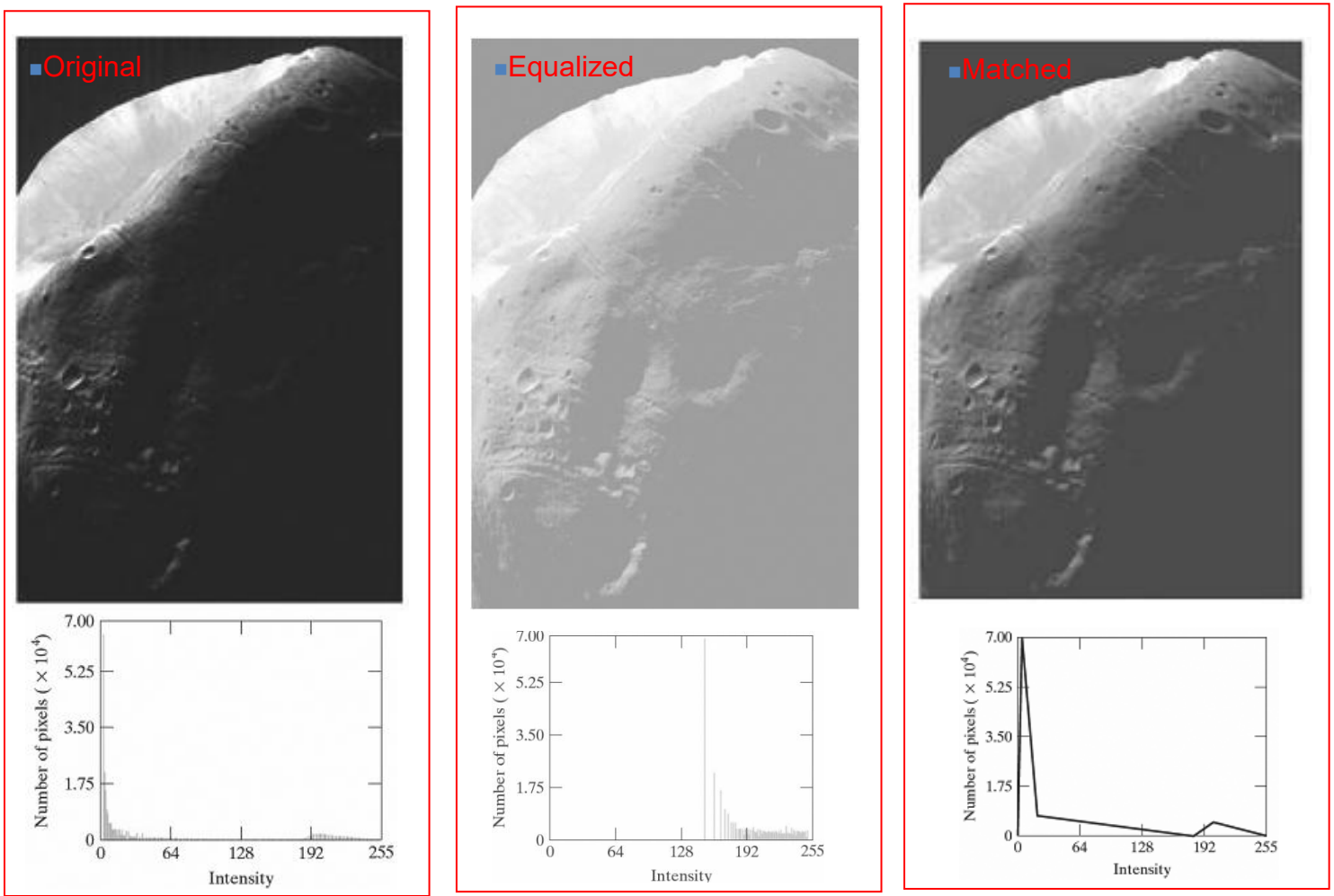


Light, washed out appearance

Histogram Matching: Comparison

- A specified target may improve the result

Specify a task-appropriate target histogram



Matching: Continuous Formulation

- Equalize the initial histogram of the image:

$$s = T(r) = (L-1)\int_0^r p_r(w)\,dw$$

- Equalize the target histogram:

$$s = G(z) = (L-1)\int_0^z p_z(w)\,dw$$

$$G(z) = T(r)$$

- Obtain the inverse transform: $z = G^{-1}(s) = G^{-1}(T(r))$

In practice, for every value of r in the image:

- get its equalized transformation $s=T(r)$.
- perform the inverse mapping $z=G^{-1}(s)$, where $s=G(z)$ is the equalized target histogram.

Matching: The Target Distribution

Consider again the 3-bit 64x64 image:

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

It is desired to transform this histogram to:

$$p_z(z_0) = 0.00 \quad p_z(z_1) = 0.00 \quad p_z(z_2) = 0.00 \quad p_z(z_3) = 0.15$$
$$p_z(z_4) = 0.20 \quad p_z(z_5) = 0.30 \quad p_z(z_6) = 0.20 \quad p_z(z_7) = 0.15$$

with $z_0 = 0, z_1 = 1, z_2 = 2, z_3 = 3, z_4 = 4, z_5 = 5, z_6 = 6, z_7 = 7.$



Matching: Quantized CDFs

The first step is to equalize the input (as before):

$$s_0 = 1,\ s_1 = 3,\ s_2 = 5,\ s_3 = 6,\ s_4 = 6,\ s_5 = 7,\ s_6 = 7,\ s_7 = 7$$

The next step is to equalize the output:

$$G(z_0) = 0 \quad G(z_1) = 0 \quad G(z_2) = 0 \quad G(z_3) = 1$$
$$G(z_4) = 2 \quad G(z_5) = 5 \quad G(z_6) = 6 \quad G(z_7) = 7$$

Notice that $G(z)$ is not strictly monotonic. We must resolve this ambiguity by choosing, e.g. the smallest value for the inverse mapping.

Matching: The Lookup Table

Perform inverse mapping: find the smallest value of z_q that is closest to s_k:

$s_k = T(r_i)$	$G(z_q)$
$s_0 = 1$	$G(z_0) = 0$
$s_1 = 3$	$G(z_1) = 0$
$s_2 = 5$	$G(z_2) = 0$
$s_3 = 6$	$G(z_3) = 1$
$s_4 = 6$	$G(z_4) = 2$
$s_5 = 7$	$G(z_5) = 5$
$s_6 = 7$	$G(z_6) = 6$
$s_7 = 7$	$G(z_7) = 7$

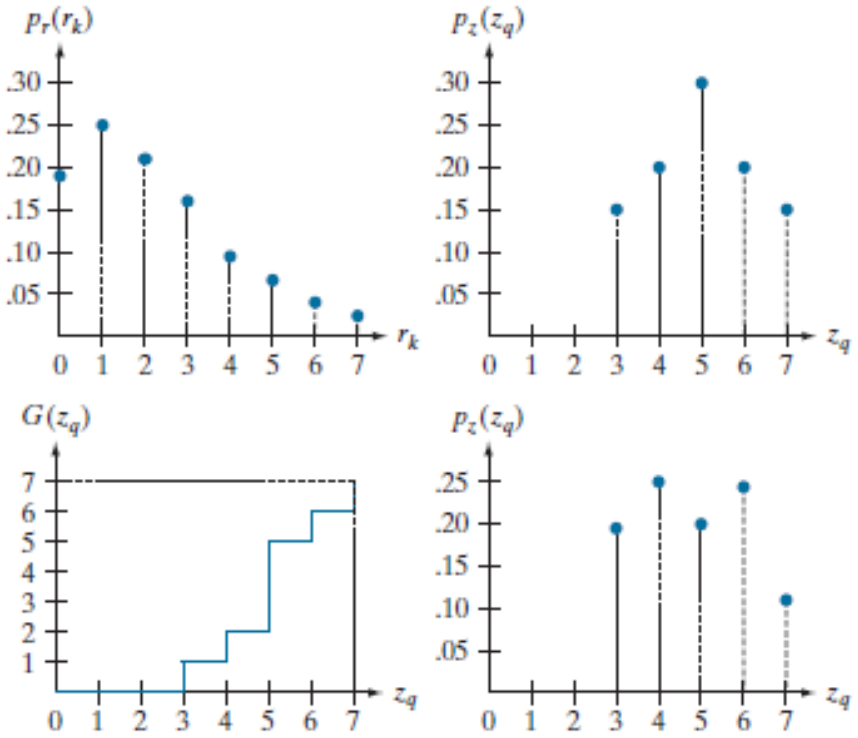


$s_k \rightarrow z_q$
$1 \rightarrow 3$
$3 \rightarrow 4$
$5 \rightarrow 5$
$6 \rightarrow 6$
$7 \rightarrow 7$

e.g. every pixel with value $s_0=1$ in the histogram-equalized image would have a value of 3 (z_3) in the histogram-specified image.

Matching: Discrete Limitations

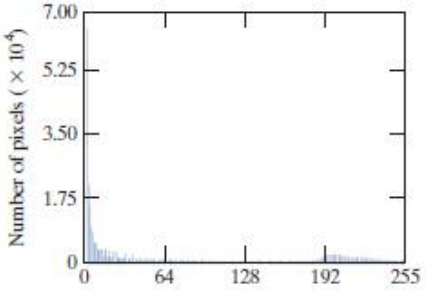
- Notice that due to discretization, the resulting histogram will rarely be exactly the same as the desired histogram.



Matching: Image and Histogram

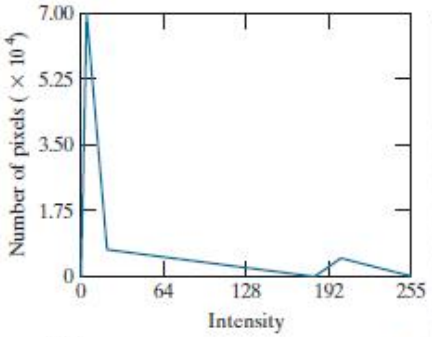
Original





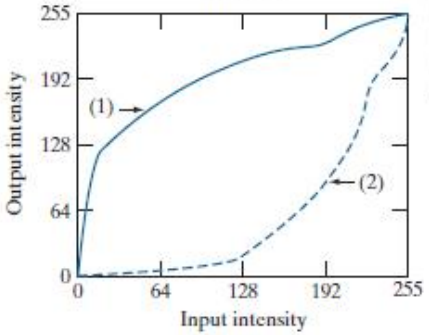
Histogram Specification

Specified Histogram

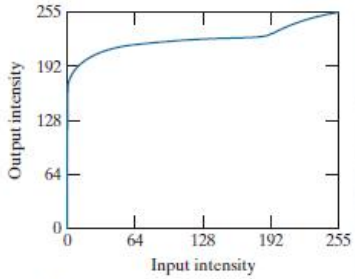




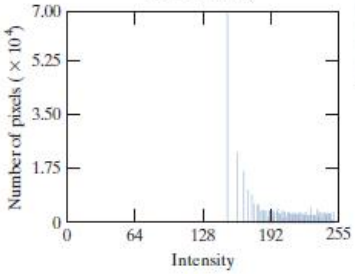
Transformation function and its inverse



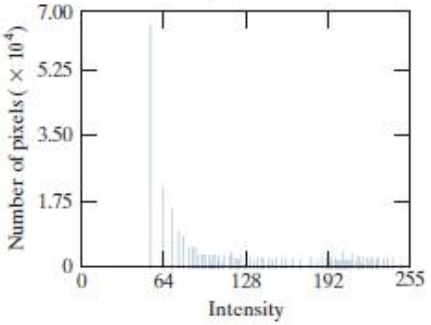
Histogram Equalization







Resulting histogram



Global and Local Equalization

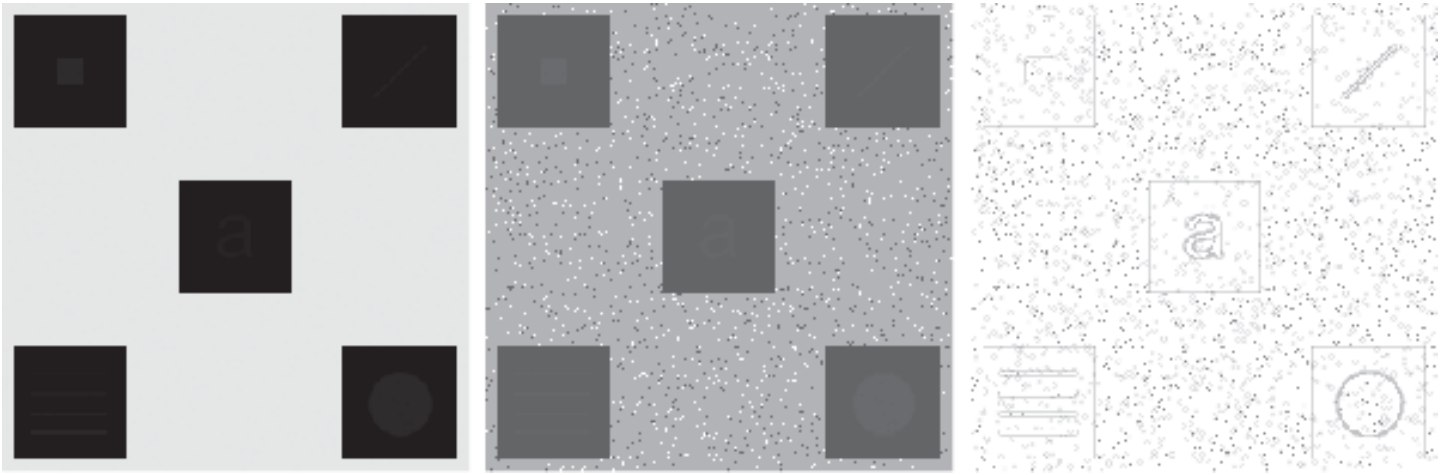


FIGURE 3.26 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization applied to (a), using a neighborhood of size 3 × 3.

- Image in (a) is slightly noisy but the noise is imperceptible.
- HE enhances the noise in smooth regions (b).
- Local HE reveals structures having values close to the values of the squares and small sizes to influence HE (c).

Local Histogram Processing

- Choose a neighborhood around each pixel.
- Estimate its local histogram and CDF.
- Use the local mapping to update the center pixel.
- Window size affects local adaptation and noise sensitivity.



Same Pixel, Different Contexts

Dark neighborhood

0.1	**0.1**	**0.1**
0.1	0.4	0.1
0.1	0.1	0.1

Bright neighborhood

0.8	**0.8**	**0.8**
0.8	0.4	0.8
0.8	0.8	0.8

The center pixels have the same input intensity.
Must a global point mapping give them the same output?
What additional information can a local method use?

Local Mean and Standard Deviation

Region	Pixel values	Mean	Std. deviation
A	0.05, 0.15, 0.05, 0.15	0.10	0.05
B	0.80, 0.80, 0.80, 0.80	0.80	0.00

- Local mean describes average intensity.
- Local standard deviation describes intensity variation.
- Intensity variation may come from structures or noise.

在图像增强中使用直方图统计

如何实现?

1. 如何分辨暗区域和亮区域?

均值是平均灰度度量，因此考虑局部均值同全局均值比较，用于区分暗区域和亮区域。
设m_G 是全局均值，则：
若$m_{S_{xy}} \leq k_0 m_G, k_0$是小于1.0的正常数，则就认为点(x,y)是需要处理的候选点。

2. 如何增强对比度低的区域？找到需要处理的对比度低的区域?

方差是灰度对比度的度量，因此考虑局部标准差同全局标准差比较，用于区分高对比度和低对比度区域。设σ_G 是全局标准差，则：
若$\sigma_{S_{xy}} \leq k_2 \sigma_G, k_2$是正常数，则就认为点(x,y)是需要处理的候选点。
当我们关注的是增强亮区域，则k_2大于1；若关注暗区域，则k_2小于1。

3. 如何避免增强标准差为零的恒定区域?

限制能够接受的最低的对比度值：
若$k_1 \sigma_G \leq \sigma_{S_{xy}}, k_1$是常数，对局部标准差设置一个较低限制值，**避免**增强标准差为零的恒定区域。

Contrast-Limited Local Enhancement

- CLAHE （Contrast Limited Adaptive Histogram Equalization) uses local histograms with a contrast limit.
- Histogram clipping limits excessive local amplification.
- Interpolation reduces discontinuities between local regions.
- Check both visible detail and noise when choosing parameters.

Choosing an Enhancement Method

Observed problem	Possible method	Check
Dark detail is hard to see	Gamma or log	Bright-region compression
Narrow intensity range	Stretching or HE	Clipping and merged levels
A target distribution is needed	Histogram matching	Suitability of the target
Small regions are overlooked	Local processing	Noise and local artifacts

Select a method from the task and the image characteristics.

Review Questions

- Does making an image brighter always improve contrast?
- Can two different images have the same histogram?
- Why is a discrete equalized histogram usually not flat?
- Why can local enhancement reveal both detail and noise?